

The geometric progression

Multiply by the same number every time, and growth stops being steady and becomes explosive.

LESSON 79–80

2 × 40 MIN

ALGEBRA 9, §32

IGX 11.5

LESSON CLOCK

16'

20'

22'

18'

4'

The definition	16 min
The nth term	20 min
The middle-term property	22 min
Growth and decay	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define a geometric progression and find its common ratio.
- Use $b_n = b_1q^{n-1}$ in all four directions.
- Use the characteristic property $b_n^2 = b_{n-1} b_{n+1}$.
- Insert geometric means and recognise a GP in a word problem.

TEXTBOOK

National	pp. 188–194
Cambridge	Core & Extended, pp. 241–247
Workbook	Exercise 11.5

01

Explanation

The definition

A sequence is a **geometric progression** if each term after the first is obtained by multiplying by the same non-zero number q , the **common ratio**.

$$b_{n+1} = b_n q \text{ equivalently } q = \frac{b_{n+1}}{b_n}$$

SEQUENCE	B_1	Q	BEHAVIOUR
2, 6, 18, 54, ...	2	3	grows fast
16, 8, 4, 2, ...	16	$\frac{1}{2}$	decays to zero
1, -2, 4, -8, ...	1	-2	alternates and grows
5, 5, 5, ...	5	1	constant
1, 3, 6, 10, ...	1	—	not a GP

NEITHER B_1 NOR Q MAY BE ZERO

A zero term would make the next ratio $\frac{0}{0}$ and destroy the definition. This is the one structural difference from an arithmetic progression, where $d = 0$ is perfectly allowed.

The n th term

Starting at b_1 and multiplying by q a total of $n - 1$ times:

$$b_n = b_1 q^{n-1}$$

GIVEN	FIND	HOW
b_1, q, n	b_n	substitute
b_1, q, b_n	n	compare powers
two terms	b_1, q	divide one equation by the other

The last row is the standard technique, and it is why a GP question is often **easier** than the corresponding AP one: dividing $b_5 = b_1q^4$ by $b_2 = b_1q$ gives q^3 at once, with b_1 eliminated.

Example. $b_2 = 6$, $b_5 = 162$. Then $q^3 = 27$, $q = 3$, and $b_1 = 2$.

THE EXPONENT IS $N - 1$ AGAIN

Exactly as in an AP, the first term needs no step at all. Writing b_1q^n is the single commonest error of the chapter — and it is caught instantly by testing $n = 1$.

The middle-term property

Each term is the geometric mean of its neighbours, in absolute value:

$$b_n^2 = b_{n-1} \cdot b_{n+1}$$

It is the multiplicative twin of $2a_n = a_{n-1} + a_{n+1}$, and it is the standard way to test whether three numbers form a GP or to fill a gap.

QUESTION	WORKING	ANSWER
3, x, 27	$x^2 = 81$	$x = \pm 9$
x, 6, 18	$36 = 18x$	$x = 2$
is 2, 6, 18 a GP?	$36 = 2 \cdot 18 \checkmark$	yes

THE FIRST ROW REALLY HAS TWO ANSWERS

Both 3, 9, 27 and 3, -9, 27 are geometric progressions, with $q = 3$ and $q = -3$. Giving only the positive value loses half the marks — unless the question says the terms are positive.

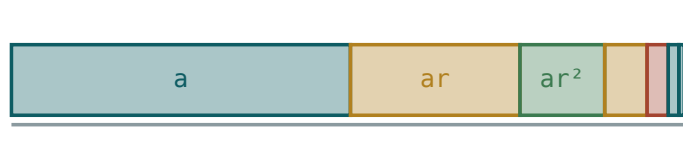
Growth and decay

The behaviour depends entirely on q .

Q	BEHAVIOUR	EXAMPLE
$q > 1$	grows without limit	a population doubling
$0 < q < 1$	decays towards zero	a bouncing ball
$q = 1$	constant	—
$q < 0$	alternates in sign	1, -2, 4, -8

$$a = 1, r = \frac{1}{2}$$

$$S = a/(1 - r)$$



the pieces never reach the wall, and never pass it

With $|q| < 1$ the terms shrink towards zero; with $|q| > 1$ they run away.

Compound interest. A sum P at $r\%$ a year is a GP with $q = 1 + \frac{r}{100}$; after n years

it is $P(1 + \frac{r}{100})^n$. Simple interest is an AP; compound interest is a GP — one

sentence that explains why the two grow so differently.

ARITHMETIC LOSES TO GEOMETRIC, ALWAYS, IN THE END

Whatever d is, and however small $q > 1$ is, the geometric progression eventually overtakes the arithmetic one. That is the whole difference between adding and multiplying, and it is why compound interest matters.

02 Worked examples

EXAMPLE 1 In a GP, $b_2 = 6$ and $b_5 = 162$. Find b_1 , q and b_7 .

$$\textcircled{1} \quad \frac{b_5}{b_2} = q^3 = \frac{162}{6} = 27$$

b_1 cancels.

$$\textcircled{2} \quad q = 3$$

$$\textcircled{3} \quad b_1 = \frac{6}{3} = 2$$

$$\textcircled{4} \quad b_7 = 2 \cdot 3^6 = 1458$$

ANSWER

$$b_1 = 2, q = 3, b_7 = 1458$$

EXAMPLE 2 Find x if 3, x , 27 is a GP.

$$\textcircled{1} \quad x^2 = 3 \cdot 27 = 81$$

The characteristic property.

$$\textcircled{2} \quad x = \pm 9$$

$$\textcircled{3} \quad 3, 9, 27 \text{ with } q = 3;$$

$$\textcircled{4} \quad 3, -9, 27 \text{ with } q = -3. \text{ Both are valid.}$$

ANSWER

$$x = \pm 9$$

EXAMPLE**3**

A ball is dropped from 2 m and rebounds to 60% of its height each bounce. How high after the 5th bounce?

① $b_1 = 2 \cdot 0.6 = 1.2$ — the first rebound.

② $q = 0.6$

③ $b_5 = 1.2 \cdot 0.6^4$

④ $= 1.2 \cdot 0.1296 \approx 0.156 \text{ m}$

ANSWER $\approx 0.16 \text{ m}$ **03 Interactive model**

Fold a sheet of paper repeatedly and count the layers; the class reaches the impossibility of eight folds and the meaning of geometric growth at the same moment.

 b_1 and q

$$b_n = 3 \cdot 2^{n-1}$$

n 1**value 3**

By the tenth term it has passed 1500. That is the difference between adding and multiplying each time.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Geometric progression	Geometrik progressiya	Геометрическая прогрессия
Common ratio	Maxraj	Знаменатель
Geometric mean	O'rta geometrik	Среднее геометрическое
Growth	O'sish	Рост
Decay	Kamayish	Убывание
Compound interest	Murakkab foiz	Сложный процент
Doubling	Ikkilanish	Удвоение
Non-zero	Noldan farqli	Отличный от нуля

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

In a GP, q equals:

$$b_{n+1} - b_n$$

$$\frac{b_{n+1}}{b_n}$$

$$b_n b_{n+1}$$

$$b_1$$

b_n equals:

$$b_1 q^n$$

$$b_1 q^{n-1}$$

$$b_1 + (n - 1)q$$

$$b_1 n q$$

2, 6, 18, ... has $b_5 =$

$$54$$

$$108$$

$$162$$

$$486$$

The characteristic property is:

$$2b_n = b_{n-1} + b_{n+1}$$

$$b_n^2 = b_{n-1} b_{n+1}$$

$$b_n = b_1 + q$$

$$b_n = nq$$

3, x , 27 gives $x =$

$$9$$

$$-9$$

$$\pm 9$$

$$15$$

Compound interest is:

an AP

a GP

neither

both

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. q of 2, 6, 18, ...

ANSWER 3

2. q of 16, 8, 4, ...

ANSWER $\frac{1}{2}$

3. q of 1, -2, 4, ...

ANSWER -2

4. $b_1 = 3, q = 2: b_5$

ANSWER 48

5. $b_1 = 5, q = 3: b_4$

ANSWER 135

6. 2, 6, 18, ...: b_5

ANSWER 162

7. Is 1, 3, 6, 10, ... a GP?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. $b_2 = 6, b_5 = 162: q$

ANSWER 3

2. Same: b_1

ANSWER 2

3. Same: b_7

ANSWER 1458

4. Find x : 3, x , 27 is a GP

ANSWER ± 9

5. Find x : x , 6, 18 is a GP

ANSWER 2

6. Which term of 2, 6, 18, ... is 486?

ANSWER The 6th

7. $b_1 = 100, q = 0.5: b_6$

ANSWER 3.125

Hard · stretch and reason

7 PROBLEMS

1. $b_3 = 12, b_6 = 96: b_1$

ANSWER 3

2. Three numbers in GP: sum 21, product 216

ANSWER 3, 6, 12

3. Insert two numbers between 2 and 54 to make a GP

ANSWER 6 and 18

4. A ball from 2 m rebounds 60%: height after 5 bounces

ANSWER ≈ 0.16 m

5. 1000 at 5% compound for 3 years

ANSWER 1157.63

6. For which x is $x, x + 2, x + 6$ a GP?

ANSWER $x = 2$

7. Show that b_1, b_3, b_5 of a GP is again a GP

ANSWER Its ratio is q^2

07 Homework

Homework — 5 tasks

Write b_1 and q at the top of every solution, and test $n = 1$ on your formula.

1. In a GP, $b_1 = 4$ and $q = 3$. Find b_6 and say which term equals 972.
2. Given $b_3 = 20$ and $b_6 = 160$, find b_1 and q .
3. Find x if 5, x , 45 is a GP.
4. Insert three numbers between 1 and 16 to make a GP.
5. A sum of 2000 earns 4% compound interest. Find its value after 5 years.

The sum of the first n terms of a geometric progression

Multiply the sum by q, subtract, and everything in the middle cancels.

LESSON 81–82

2 × 40 MIN

ALGEBRA 9, §33

IGX 11.6

LESSON CLOCK

16'

20'

20'

20'

4'

The derivation	16 min
Two arrangements	20 min
Worked problems	20 min
The chessboard	20 min
Homework	4 min

LEARNING OBJECTIVES

- Derive $S_n = b_1(q^n - 1)/(q - 1)$ by the shift-and-subtract method.
- Use the formula in the form that avoids negative numerators.
- Treat the case $q = 1$ separately.
- Apply the formula to compound interest and repayment problems.

TEXTBOOK

National	pp. 195–200
Cambridge	Core & Extended, pp. 248–253
Workbook	Exercise 11.6

01

Explanation

The derivation

Write the sum, then the same sum multiplied by q:

$$S_n = b_1 + b_1q + b_1q^2 + \dots + b_1q^{n-1}$$

$$qS_n = b_1q + b_1q^2 + \dots + b_1q^{n-1} + b_1q^n$$

Subtract the first from the second. Every term but two cancels:

$$qS_n - S_n = b_1q^n - b_1$$

$$S_n = \frac{b_1(q^n - 1)}{q - 1}, q \neq 1$$

WHY THE TRICK WORKS

Multiplying by q shifts the whole list one place along. Subtracting therefore removes everything the two lists have in common — which is all of it except the first term of one and the last of the other. The same idea will reappear in Grade 11 for series.

Two arrangements

Multiplying numerator and denominator by -1 gives an equivalent form:

$$S_n = \frac{b_1(q^n - 1)}{q - 1} = \frac{b_1(1 - q^n)}{1 - q}$$

USE	WHEN	BECAUSE
$\frac{b_1(q^n - 1)}{q - 1}$	$q > 1$	both parts are positive
$\frac{b_1(1 - q^n)}{1 - q}$	$ q < 1$	both parts are positive
$S_n = n b_1$	$q = 1$	the formula divides by zero

$Q = 1$ MUST BE CHECKED BEFORE THE FORMULA IS USED

With $q = 1$ every term is b_1 and the sum is $n b_1$ — but the formula has $q - 1$ in the denominator and is undefined. A question that sets $q = 1$ is testing exactly whether you looked.

Worked problems

PROBLEM	WORKING	ANSWER
S_6 of 2, 6, 18, ...	$\frac{2(3^6 - 1)}{2}$	728
S_5 of 3, 6, 12, ...	$\frac{3(2^5 - 1)}{1}$	93
S_4 of 16, 8, 4, ...	$\frac{16(1 - \frac{1}{16})}{1 - \frac{1}{2}}$	30

The third row is a useful check: $16 + 8 + 4 + 2 = 30$ by direct addition, which confirms the arrangement of the formula for $q < 1$.

ALWAYS SANITY-CHECK A SMALL CASE BY ADDING

Four or five terms can be added directly in a few seconds. Doing so once, at the start of the exercise, catches a mis-remembered formula before it spoils ten questions.

The chessboard

The classical problem: one grain on the first square, two on the second, four on the third, doubling to the sixty-fourth.

$$S_{64} = \frac{1(2^{64} - 1)}{2 - 1} = 2^{64} - 1 \approx 1.8 \times 10^{19}$$

That is more grain than has ever been grown. It is the standard demonstration that geometric growth is not merely “fast” but of a different order altogether.

SITUATION	THE GP	THE SUM MEANS
a bouncing ball	$b_1 = \text{first rebound}, q < 1$	total distance
annual savings with interest	$q = 1 + \frac{r}{100}$	the accumulated fund
a rumour told to 3 people each	$q = 3$	how many have heard it

02 Worked examples**EXAMPLE 1** Find S_6 for 2, 6, 18,

① $b_1 = 2, q = 3, n = 6$

② $S_6 = \frac{2(3^6 - 1)}{3 - 1}$

$q > 1$ form.

③ $3^6 = 729$

④ $S_6 = \frac{2 \cdot 728}{2} = 728$

ANSWER**728**

EXAMPLE 2 Find S_4 for 16, 8, 4,

$$(1) \quad b_1 = 16, q = \frac{1}{2}$$

$$(2) \quad S_4 = \frac{16(1 - \frac{1}{16})}{1 - \frac{1}{2}}$$

$|q| < 1$ form.

$$(3) \quad = \frac{16 \cdot \frac{15}{16}}{\frac{1}{2}} = \frac{15}{\frac{1}{2}}$$

$$(4) \quad = 30$$

Check: $16 + 8 + 4 + 2 = 30 \checkmark$

ANSWER

30

EXAMPLE 3 How many terms of 3, 6, 12, ... give a sum of 381?

$$(1) \quad b_1 = 3, q = 2$$

$$(2) \quad \frac{3(2^n - 1)}{1} = 381$$

$$(3) \quad 2^n - 1 = 127 \Rightarrow 2^n = 128$$

$$(4) \quad n = 7$$

ANSWER

$n = 7$

03 Interactive model

Do the chessboard problem on the board without the formula for the first eight squares, then with it for all sixty-four; the gap between intuition and arithmetic is the lesson.

S_n of a GP

$$S_n = 3(2^n - 1)$$

n 1

value 3

Always exactly three less than the next term, $3 \cdot 2^n$ — which is what the “multiply by q and subtract” trick leaves behind.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sum of n terms	n ta hadning yig‘indisi	Сумма n членов
Shift and subtract	Siljitib ayirish	Сдвиг и вычитание
Cancellation	Qisqarish	Сокращение
Special case	Xususiyl hol	Частный случай
Numerator	Surat	Числитель
Repayment	To‘lov	Выплата
Doubling problem	Ikkilanish masalasi	Задача об удвоении
Estimate	Baholash	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

S_n of a GP equals:

$$\frac{b_1(q^n - 1)}{q - 1}$$

$$\frac{n(b_1 + b_n)}{2}$$

$$b_1 q^n$$

$$n b_1 q$$

The formula fails when:

$$q = 0$$

$$q = 1$$

$$q < 0$$

never

When $q = 1$, S_n equals:

$$0$$

$$n b_1$$

$$b_1$$

undefined

S_6 of 2, 6, 18, ...:

S_4 of 16, 8, 4, ...:

The chessboard total is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. S_3 of 1, 2, 4

ANSWER 7

2. S_4 of 1, 3, 9, 27

ANSWER 40

3. S_5 of 3, 6, 12, ...

ANSWER 93

4. S_4 of 16, 8, 4, ...

ANSWER 30

5. S_6 of 2, 6, 18, ...

ANSWER 728

6. $q = 1, b_1 = 5, n = 8: S_8$

ANSWER 40

7. S_3 of 1, -2, 4

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. How many terms of 3, 6, 12, ... give 381?

ANSWER 7

2. $b_1 = 1, q = 2, n = 10: S_{10}$

ANSWER 1023

3. $b_1 = 5, q = 3, n = 5: S_5$

ANSWER 605

4. S_5 of 81, 27, 9, ...

ANSWER 121

5. $b_2 = 6, b_5 = 162: S_5$

ANSWER 242

6. Sum of $1 + 2 + 4 + \dots + 2^{63}$

ANSWER $2^{64} - 1$

7. S_4 of 2, -6, 18, -54

ANSWER -40

Hard · stretch and reason

7 PROBLEMS

1. How many terms of $1, 2, 4, \dots$ give 1023 ?

ANSWER 10

2. $S_3 = 21$ and $q = 2$: find b_1

ANSWER 3

3. A ball from 2 m rebounds 50% : total distance to rest

ANSWER 6 m

4. Save 1000 at the start of each year at 10% for 5 years

ANSWER ≈ 6716

5. $S_4 = 30$ and $S_8 = 510$: find q

ANSWER 2

6. Show $S_n = \frac{b_n q - b_1}{q - 1}$

ANSWER Substitute $b_n = b_1 q^{n-1}$

7. A rumour: each teller tells 3 new people. After 5 rounds, how many know?

ANSWER 364

07 Homework

Homework — 5 tasks

Check every answer by adding the first three or four terms directly.

- Find S_5 for $2, 8, 32, \dots$
- Find S_5 for $243, 81, 27, \dots$
- How many terms of $1, 3, 9, \dots$ give a sum of 364 ?
- A GP has $b_1 = 6$ and $q = \frac{1}{3}$. Find S_4 .

5. Explain why the formula cannot be used when $q = 1$, and give the sum in that case.

The infinite decreasing geometric progression

Add infinitely many numbers and get a finite answer — provided each is small enough.

LESSON 83–84

2 × 40 MIN

ALGEBRA 9, §34

IGX 11.7

LESSON CLOCK

15'

19'

22'

20'

4'

When infinitely many terms have a sum	15 min
The formula	19 min
Recurring decimals	22 min
Distance problems	20 min
Homework	4 min

LEARNING OBJECTIVES

- State the condition $|q| < 1$ for an infinite geometric sum to exist.
- Use $S = b_1/(1 - q)$ and explain where it comes from.
- Convert a recurring decimal into a fraction by summing a GP.
- Solve problems about total distance and repeated halving.

TEXTBOOK

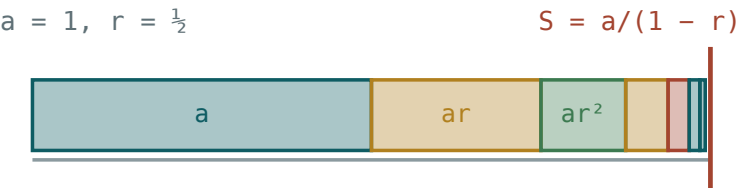
National	pp. 201–206
Cambridge	Core & Extended, pp. 254–258
Workbook	Exercise 11.7

01

Explanation

When infinitely many terms have a sum

Consider $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$. The partial sums are 1, 1.5, 1.75, 1.875, ... — they climb, but never past 2.



the pieces never reach the wall, and never pass it

The partial sums of a decreasing GP creep towards a ceiling they never reach.

Compare $1 + 2 + 4 + 8 + \dots$, whose partial sums are $1, 3, 7, 15, \dots$ and run away to infinity. The difference is entirely in q .

$ Q $	Q^n	THE SUM
$ q < 1$	shrinks to 0	exists — the progression converges
$ q \geq 1$	does not shrink	does not exist — it diverges

“INFINITELY MANY” DOES NOT MEAN “INFINITELY LARGE”

This is the first place in school mathematics where an endless process has a finite answer, and it is worth pausing on. The terms must shrink fast enough — and geometric shrinking, by a constant factor each time, is fast enough.

The formula

In $S_n = \frac{b_1(1 - q^n)}{1 - q}$, if $|q| < 1$ then q^n becomes as small as you like as n grows.

Dropping it:

$$S = \frac{b_1}{1 - q}, |q| < 1$$

PROGRESSION	B_1	Q	S
$1, \frac{1}{2}, \frac{1}{4}, \dots$	1	$\frac{1}{2}$	2
$9, 3, 1, \dots$	9	$\frac{1}{3}$	$\frac{27}{2}$
$4, -2, 1, \dots$	4	$-\frac{1}{2}$	$\frac{8}{3}$
$1, 2, 4, \dots$	1	2	none

CHECK $|Q| < 1$ BEFORE WRITING THE FORMULA DOWN

Applying $\frac{b_1}{1 - q}$ to $1 + 2 + 4 + \dots$ gives $\frac{1}{-1} = -1$ — a finite negative answer to a sum of positive numbers. The formula is not merely inaccurate outside its condition; it is meaningless there.

Recurring decimals

Every recurring decimal is an infinite GP in disguise, which is why every one of them is a fraction.

Example. $0.3333... = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + ...$ — a GP with $b_1 = \frac{3}{10}$ and $q = \frac{1}{10}$:

$$S = \frac{\frac{3}{10}}{1 - \frac{1}{10}} = \frac{\frac{3}{10}}{\frac{9}{10}} = \frac{1}{3}$$

DECIMAL	B_1	Q	FRACTION
0.333...	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{3}$
0.777...	$\frac{7}{10}$	$\frac{1}{10}$	$\frac{7}{9}$
0.121212...	$\frac{12}{100}$	$\frac{1}{100}$	$\frac{4}{33}$
0.999...	$\frac{9}{10}$	$\frac{1}{10}$	1

0.999... = 1, AND THIS IS THE PROOF

Not “nearly 1”, not “1 for practical purposes” — exactly 1. The two symbols name the same number, and the geometric sum above is the shortest honest demonstration of it.

Distance problems

A bouncing ball. Dropped from h and rebounding to a fraction k of its height, the total distance travelled before coming to rest is

$$h + 2hk + 2hk^2 + ... = h + \frac{2hk}{1 - k}$$

The first drop is counted once; every later height is travelled twice, up and down. Forgetting the factor 2, or double-counting the first drop, are the two standard errors.

PROBLEM	TOTAL
dropped from 2 m, rebound 50%	$2 + \frac{2 \cdot 2 \cdot 0.5}{0.5} = 6 \text{ m}$
a square of side 1, each next joining midpoints	perimeters sum to $\frac{4}{1 - \frac{\sqrt{2}}{2}}$
a pendulum swinging 80% of the previous arc, first arc 40 cm	200 cm

02 Worked examples

EXAMPLE 1 Find the sum of $9 + 3 + 1 + \frac{1}{3} + \dots$

$$(1) \quad b_1 = 9, q = \frac{1}{3}$$

$$(2) \quad |q| < 1 \checkmark \text{ — the sum exists.}$$

$$(3) \quad S = \frac{9}{1 - \frac{1}{3}} = \frac{9}{\frac{2}{3}}$$

$$(4) \quad = \frac{27}{2} = 13.5$$

ANSWER

$$\frac{27}{2}$$

EXAMPLE 2 Write $0.121212\dots$ as a fraction.

$$\textcircled{1} \quad b_1 = \frac{12}{100}, q = \frac{1}{100}$$

$$\textcircled{2} \quad S = \frac{\frac{12}{100}}{1 - \frac{1}{100}}$$

$$\textcircled{3} \quad = \frac{12}{99}$$

$$\textcircled{4} \quad = \frac{4}{33}$$

Cancel by 3.

ANSWER

$$\frac{4}{33}$$

EXAMPLE 3 A ball is dropped from 2 m and rebounds to half its height each time. Find the total distance travelled.

$\textcircled{1}$ The first drop: 2 m.
Counted once.

$\textcircled{2}$ Then up and down 1, 0.5, 0.25, ..., each twice.

$$\textcircled{3} \quad 2 \cdot \frac{1}{1 - 0.5} = 4$$

$$\textcircled{4} \quad 2 + 4 = 6 \text{ m}$$

ANSWER

6 m

03 Interactive model

Walk half the distance to the wall, then half of what remains, and again; the class sees an endless process with a finite destination before any algebra is written.

Partial sums approaching a ceiling

$$S_n = 8(1 - 0.5^n)$$

n 1

value 4

The terms are 4, 2, 1, $\frac{1}{2}$, ... and the sum never reaches 8 — but it gets as close as you like.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Infinite progression	Cheksiz progressiya	Бесконечная прогрессия
Decreasing	Kamayuvchi	Убывающая
Sum of the progression	Progressiya yig'indisi	Сумма прогрессии
Converges	Yaqinlashadi	Сходится
Diverges	Uzoqlashadi	Расходится
Recurring decimal	Davriy o'nli kasr	Периодическая дробь
Limit	Limit	Предел
Condition	Shart	Условие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

An infinite GP has a sum when:

$$q > 0$$

$$|q| < 1$$

$$q > 1$$

always

S equals:

$$\frac{b_1}{1 - q}$$

$$\frac{b_1}{1 + q}$$

$$\frac{1 - q}{b_1}$$

$$b_1 q$$

$1 + \frac{1}{2} + \frac{1}{4} + \dots$ equals:

$$1.5$$

$$2$$

$$\infty$$

$$2.5$$

$0.999\dots$ equals:

nearly 1

1

0.99

$\frac{9}{10}$

$1 + 2 + 4 + \dots$ has:

-1

no sum

∞ as an answer

2

$0.333\dots$ equals:

$\frac{1}{3}$

$\frac{3}{10}$

$\frac{33}{100}$

$\frac{1}{9}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up
7 PROBLEMS

1. $1 + \frac{1}{2} + \frac{1}{4} + \dots$

ANSWER 2

2. $9 + 3 + 1 + \dots$

 ANSWER $\frac{27}{2}$

3. $b_1 = 6, q = \frac{1}{3} : S$

ANSWER 9

4. $b_1 = 4, q = -\frac{1}{2} : S$

 ANSWER $\frac{8}{3}$

5. Does $1 + 2 + 4 + \dots$ have a sum?

ANSWER No

6. $0.333\dots$ as a fraction

 ANSWER $\frac{1}{3}$

7. $0.777\dots$ as a fraction

 ANSWER $\frac{7}{9}$
Medium · the standard question
7 PROBLEMS

1. $0.121212\dots$ as a fraction

ANSWER $\frac{4}{33}$

2. $0.999\dots$

ANSWER 1

3. $b_1 = 12, q = 0.25$: S

ANSWER 16

4. $S = 8$ and $q = \frac{1}{2}$: find b_1

ANSWER 4

5. $S = 27$ and $b_1 = 18$: find q

ANSWER $\frac{1}{3}$

6. Ball from 2 m , rebound 50% : total distance

ANSWER 6 m

7. Pendulum, first arc 40 cm , each 80% : total

ANSWER 200 cm

Hard · stretch and reason

7 PROBLEMS

1. $0.4545\dots$ as a fraction

ANSWER $\frac{5}{11}$

2. $0.1666\dots$ as a fraction

ANSWER $\frac{1}{6}$

3. Ball from 10 m , rebound $\frac{2}{3}$: total distance

ANSWER 50 m

4. An infinite GP has $S = 9$ and $b_1 = 6$: find b_3

ANSWER $\frac{2}{3}$

5. For which x does $1 + x + x^2 + \dots$ have a sum?

ANSWER $-1 < x < 1$

6. Sum that progression when $x = \frac{1}{4}$

ANSWER $\frac{4}{3}$

7. Squares: side 1 , each next joining midpoints. Sum of the areas

ANSWER 2

07 Homework

Homework — 5 tasks

State $|q| < 1$ explicitly before using the formula, every time.

- Find the sum of $16 + 4 + 1 + \frac{1}{4} + \dots$
- Write $0.636363\dots$ as a fraction.

3. An infinite GP has $S = 12$ and $q = \frac{2}{3}$. Find b_1 .
4. A ball dropped from 3 m rebounds to $\frac{1}{3}$ of its height. Find the total distance.
5. Explain why $2 + 4 + 8 + \dots$ has no sum.

Chapter exercises — Cambridge sequences and the nth-term test

The same progressions, in the language and the question style of an IGCSE paper.

LESSON 85

1 × 40 MIN

ALGEBRA 9, IV BOB MASHQLARI

IGX 11 REVIEW

LESSON CLOCK

7'

11'

13'

7'

2'

Which type is it?	7 min
Linear and geometric	11 min
Quadratic sequences	13 min
Exam phrasing	7 min
Homework	2 min

LEARNING OBJECTIVES

- Find the nth term of a linear, a quadratic and a geometric sequence.
- Use the second-difference method on a quadratic sequence.
- Answer in the phrasing Cambridge expects: “write down an expression for the nth term”.
- Decide which type a sequence is before choosing a method.

TEXTBOOK

National	pp. 207–210
Cambridge	Core & Extended, pp. 259–264
Workbook	Exercise 11 review

01

Explanation

Which type is it?

TEST	IF CONSTANT	TYPE	NTH TERM
first differences	d	linear (AP)	$dn + c$
second differences	$2a$	quadratic	$an^2 + bn + c$
ratios	q	geometric (GP)	b_1q^{n-1}

Run the three tests in that order. It takes fifteen seconds and it decides everything that follows.

CAMBRIDGE ASKS FOR THE NTH TERM FAR MORE OFTEN THAN FOR A PARTICULAR TERM

An Uzbek paper says “find a_{20} ”; an IGCSE paper says “write down an expression for the n th term”. The second is harder and is worth practising directly.

Linear and geometric

Linear. 5, 8, 11, 14, ... has $d = 3$, so the n th term is $3n$ plus a correction. At $n = 1$, $3 \cdot 1 = 3$ but the term is 5, so add 2: the answer is $3n + 2$.

Geometric. 3, 6, 12, 24, ... has $q = 2$, so the n th term is $3 \cdot 2^{n-1}$.

SEQUENCE	NTH TERM
5, 8, 11, 14, ...	$3n + 2$
20, 17, 14, ...	$23 - 3n$
3, 6, 12, 24, ...	$3 \cdot 2^{n-1}$
2, -6, 18, ...	$2(-3)^{n-1}$

ALWAYS TEST YOUR EXPRESSION AT $N = 1$

It costs two seconds and catches the “off by d ” error that this topic generates more than any other.

Quadratic sequences

If the first differences are not constant but the **second** differences are, the sequence is quadratic.

Example. 3, 8, 15, 24, 35, ...

TERMS	3	8	15	24	35
first differences		5	7	9	11
second differences			2	2	2

The second difference is $2a$, so $a = 1$. Subtracting n^2 from the terms leaves 2, 4, 6, 8, 10 — which is $2n$. So the n th term is $n^2 + 2n$, and a check at $n = 1$ gives 3 ✓.

THE METHOD IN ONE LINE

Halve the second difference to get a ; subtract an^2 from every term; what remains is linear, and you already know how to do that.

Exam phrasing

CAMBRIDGE SAYS	YOU MUST GIVE
“write down the next two terms”	two numbers
“write down an expression for the n th term”	a formula in n
“show that the n th term is...”	the working, not just the answer
“explain why 50 is not a term”	a sentence with a reason

“EXPLAIN” AND “SHOW THAT” CARRY THE MARKS IN THE WORDS

A correct number with no sentence scores zero on an “explain” question. “Because $3n + 2 = 50$ gives $n = 16$, which is a whole number, 50 **is** a term” — that is the answer, including the reason.

02 Worked examples

EXAMPLE 1 Write down an expression for the n th term of 5, 8, 11, 14,

- 1

First differences: all 3 — linear.
- 2

Start from $3n$.
- 3

At $n = 1$: 3, but the term is 5.
Add 2.
- 4

$3n + 2$
Check $n = 4$: 14 ✓

ANSWER

$3n + 2$

EXAMPLE 2 Find the n th term of 3, 8, 15, 24, 35,

- ① First differences 5, 7, 9, 11; second differences all 2.
Quadratic.

② $2a = 2 \Rightarrow a = 1.$

- ③ Subtract n^2 : 2, 4, 6, 8, 10 — that is $2n$.

④ $n^2 + 2n$
Check $n = 1$: 3 ✓

ANSWER

$n^2 + 2n$

EXAMPLE 3 Explain why 50 is not a term of 2, 6, 18, 54,

- ① Ratios all 3 — geometric, n th term $2 \cdot 3^{n-1}$.

② Solve $2 \cdot 3^{n-1} = 50$.

③ $3^{n-1} = 25$ — not a power of 3.

- ④ So there is no whole n , and 50 is not a term.
The sentence is the answer.

ANSWER

Not a term — 25 is not a power of 3

03 Interactive model

Give one linear, one quadratic and one geometric sequence unlabelled and let the class sort them by running the three tests; the sorting is the skill being taught.

Which type, and what is the n th term?

3, 7, 11, 15, ... is:

arithmetic

geometric

quadratic

none of these

Its n th term is:

$4n$

$4n - 1$

$4n + 3$

$3n + 4$

2, 6, 18, 54, ... is:

arithmetic

geometric

quadratic

none

Its n th term is:

$$2 \cdot 3^n$$

$$2 \cdot 3^{n-1}$$

$$3 \cdot 2^n$$

$$6^n$$

1, 4, 9, 16, ... has second differences:

$$0$$

$$1$$

$$2$$

$$3$$

Its n th term is:

$$n^2$$

$$2n$$

$$n^2 + 1$$

$$2n - 1$$

A constant first difference means:

geometric

arithmetic

quadratic

random

A constant second difference means the n th term is:

linear

quadratic

geometric

constant

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Linear sequence	Chiziqli ketma-ketlik	Линейная последовательность
Quadratic sequence	Kvadratlik ketma-ketlik	Квадратичная последовательность
First difference	Birinchii ayirma	Первая разность
Second difference	Ikkinchi ayirma	Вторая разность
Expression	Ifoda	Выражение
Position	O'rin	Позиция
Pattern	Naqsh	Закономерность
Justify	Asoslash	Обосновать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Constant first differences mean:

Constant second differences mean:

Constant ratios mean:

The n th term of 5, 8, 11, ...:

$$3n$$

$$3n + 2$$

$$3n - 2$$

$$5n$$

Second difference 2 gives $a =$

$$1$$

$$2$$

$$4$$

$$\frac{1}{2}$$

An “explain” question needs:

a number

a sentence with a reason

a graph

a table

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. n th term of 2, 4, 6, 8, ...

ANSWER $2n$

2. n th term of 5, 8, 11, ...

ANSWER $3n + 2$

3. n th term of 20, 17, 14, ...

ANSWER $23 - 3n$

4. n th term of 3, 6, 12, 24, ...

ANSWER $3 \cdot 2^{n-1}$

5. Next two terms of 1, 4, 9, 16, ...

ANSWER 25, 36

6. Type of 2, 6, 18, 54, ...

ANSWER Geometric

7. Type of 1, 3, 6, 10, ...

ANSWER Quadratic

Medium · the standard question

7 PROBLEMS

1. n th term of 3, 8, 15, 24, ...

ANSWER $n^2 + 2n$

2. n th term of 1, 4, 9, 16, ...

ANSWER n^2

3. n th term of 2, 5, 10, 17, ...

ANSWER $n^2 + 1$

4. n th term of 2, -6, 18, ...

ANSWER $2(-3)^{n-1}$

5. Is 50 a term of 2, 6, 18, ...?

ANSWER No

6. Is 50 a term of 5, 8, 11, ...?

ANSWER Yes — the 16th

7. n th term of 1, 3, 6, 10, ...

ANSWER $\frac{n(n+1)}{2}$

Hard · stretch and reason

7 PROBLEMS

1. n th term of 4, 11, 22, 37, ...

ANSWER $2n^2 + n + 1$

2. n th term of 0, 3, 8, 15, ...

ANSWER $n^2 - 1$

3. n th term of 5, 11, 21, 35, ...

ANSWER $2n^2 + 3$

4. Which term of $n^2 + 2n$ equals 120?

ANSWER The 10th

5. Explain why 100 is not a term of $n^2 + 2n$

ANSWER $n^2 + 2n = 100$ has no whole root

6. n th term of 1, 2, 4, 8, ... and of 1, 2, 4, 7, ...

ANSWER 2^{n-1} and $\frac{n^2 - n + 2}{2}$

7. A sequence has n th term $n^2 - 10n$. Its least term

ANSWER -25, at $n = 5$

07 Homework

Homework — 5 tasks

Test every expression at $n = 1$ and at $n = 3$ before writing it down as final.

- Write an expression for the n th term of 7, 12, 17, 22,
- Write an expression for the n th term of 2, 8, 18, 32,
- Write an expression for the n th term of 5, 15, 45,
- Explain why 60 is not a term of 7, 12, 17,

5. Find the n th term of 6, 11, 18, 27,

Control work 7, and progressions closed

Both progressions, both sums, and the infinite case — Chapter IV in one paper.

LESSON 86–87

2 × 40 MIN

ALGEBRA 9, NAZORAT ISHI 7

IGX 11 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions

3 min

The paper

40 min

Answers

12 min

Diagnosis and rewrite

20 min

The map

5 min

LEARNING OBJECTIVES

• Find a term and a sum of an arithmetic and of a geometric progression under time.

• Sum an infinite decreasing progression, stating the condition.

• Find the n th term of a sequence given only its first few terms.

• Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National

pp. 167–210

Cambridge

Core & Extended, pp. 220–264

Workbook

Control paper A7

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	In an AP, $a_1 = 4$, $d = 6$: find a_{12} and S_{12}	5	L72–76
2	In a GP, $b_2 = 12$ and $b_5 = 96$: find b_1 and q	5	L79–80
3	Find S_6 of 3, 6, 12, ...	5	L81–82
4	Sum $12 + 4 + \frac{4}{3} + \dots$ to infinity	5	L83–84
5	Write 0.272727... as a fraction	5	L83–84
6	Find the n th term of 2, 5, 10, 17, ...	5	L85

WHERE THE MARKS ACTUALLY GO

Q2 carries two marks for dividing rather than solving two equations; Q4 one for stating $|q| < 1$; Q5 two for identifying b_1 and q correctly; Q6 one for running the second-difference test before guessing.

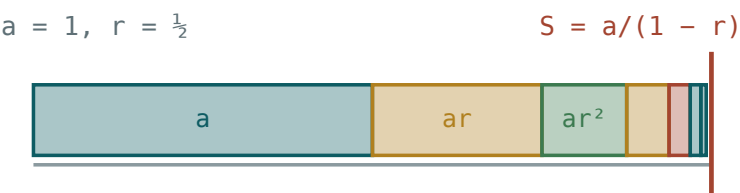
Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
exponent n instead of $n - 1$	$b_1 q^n$	test $n = 1$
AP formula used on a GP	$a_1 + (n - 1)d$	check the ratios first
the two S_n forms mixed	$\frac{b_1(q^n - 1)}{1 - q}$	pick one and write it out
condition not stated	$S = \frac{b_1}{1 - q}$ <i>with no comment</i>	“since $ q < 1 \dots$ ”
recurring decimal b_1 wrong	$b_1 = 27$	$b_1 = \frac{27}{100}$
fraction not cancelled	$\frac{27}{99}$	$\frac{3}{11}$
quadratic sequence guessed	$3n - 1$	second differences first

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter IV as one map

	ARITHMETIC	GEOMETRIC
step	add d	multiply by q
n	$a_n = a_1 + (n - 1)d$	$b_n = b_1q^{n-1}$
middle term	$2a_n = a_{n-1} + a_{n+1}$	$b_n^2 = b_{n-1} b_{n+1}$
sum	$\frac{n(a_1 + a_n)}{2}$	$\frac{b_1(q^n - 1)}{q - 1}$
proof of the sum	pair the ends	shift and subtract
infinite sum	never	$\frac{b_1}{1 - q}, q < 1$
in life	simple interest	compound interest



the pieces never reach the wall, and never pass it

The geometric case, where the partial sums have a ceiling — the arithmetic ones never do.

LOOKING FORWARD

The last chapter of the year is probability: events, the classical definition, and relative frequency. It borrows the counting habits of this chapter but nothing of its formulae — and it finishes the Grade 9 course.

02 Worked examples

EXAMPLE 1 Model answer, Q2: $b_2 = 12, b_5 = 96$.

$$\textcircled{1} \quad \frac{b_5}{b_2} = q^3 = \frac{96}{12} = 8$$

b_1 cancels.

$$\textcircled{2} \quad q = 2$$

$$\textcircled{3} \quad b_1 = \frac{12}{2} = 6$$

$$\textcircled{4} \quad b_1 = 6, q = 2$$

ANSWER

$$b_1 = 6, q = 2$$

EXAMPLE 2 Model answer, Q4: sum $12 + 4 + \frac{4}{3} + \dots$

$$\textcircled{1} \quad q = \frac{4}{12} = \frac{1}{3}$$

$$\textcircled{2} \quad |q| < 1, \text{ so the sum exists.}$$

State it.

$$\textcircled{3} \quad S = \frac{12}{1 - \frac{1}{3}} = \frac{12}{\frac{2}{3}}$$

$$\textcircled{4} \quad = 18$$

ANSWER

$$18$$

EXAMPLE 3 Model answer, Q5: write $0.272727\dots$ as a fraction.

$$① \quad b_1 = \frac{27}{100}, q = \frac{1}{100}$$

$$② \quad S = \frac{\frac{27}{100}}{1 - \frac{1}{100}}$$

$$③ \quad = \frac{27}{99}$$

$$④ \quad = \frac{3}{11}$$

Cancel by 9.

ANSWER

$$\frac{3}{11}$$

03 Interactive model

Put an AP and a GP side by side with the same first term and ask which is larger at $n = 5$ and at $n = 20$; the answer changes, and that is the chapter in one picture.

Chapter IV in twelve questions

In an AP, d is:

$$\frac{a_{n+1}}{a_n}$$

$$a_{n+1} - a_n$$

$$a_1$$

$$n$$

In a GP, q is:

$$b_{n+1} - b_n$$

$$\frac{b_{n+1}}{b_n}$$

$$b_1$$

$$n$$

a_n equals:

$$a_1 + nd$$

$$a_1 + (n - 1)d$$

$$nd$$

$$a_1 d$$

b_n equals:

$$b_1 q^n$$

$$b_1 q^{n-1}$$

$$b_1 + nq$$

$$nq$$

 S_n of an AP:

$$\frac{n(a_1 + a_n)}{2}$$

$$\frac{b_1(q^n - 1)}{q - 1}$$

$$n a_1$$

$$a_1 + a_n$$

 S_n of a GP:

$$\frac{n(b_1 + b_n)}{2}$$

$$\frac{b_1(q^n - 1)}{q - 1}$$

$$n b_1$$

$$b_1 q^n$$

The GP sum formula fails when:

$$q = 0$$

$$q = 1$$

$$q < 0$$

never

An infinite GP has a sum when:

$$q > 0$$

$$|q| < 1$$

$$q > 1$$

always

$0.999\dots$ equals:

nearly 1

$$1$$

$$0.99$$

$$\frac{9}{10}$$

Constant second differences mean:

linear

quadratic

geometric

none

Compound interest is:

an AP

a GP

neither

both

The sum of the first n odd numbers:

n

n^2

$2n$

$n(n+1)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Arithmetic progression	Arifmetik progressiya	Арифметическая прогрессия
Geometric progression	Geometrik progressiya	Геометрическая прогрессия
Common ratio	Maxraj	Знаменатель
Infinite sum	Cheksiz yig'indi	Бесконечная сумма
Condition	Shart	Условие
nth term	n-had	n-й член
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q2 is fastest by:

two equations

dividing one by the other

a graph

guessing

Q4 must include:

a decimal

the condition $|q| < 1$

a graph

 n In Q5, b_1 is:

27

 $\frac{27}{100}$ $\frac{27}{10}$

0.27

Q6 begins with:

guessing

the second differences

the ratios

a graph

2, 5, 10, 17, ... has n th term:

$3n - 1$

$n^2 + 1$

$2n^2$

$n^2 + n$

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $a_1 = 4, d = 6: a_{12}$

ANSWER 70

2. Same: S_{12}

ANSWER 444

3. $b_2 = 12, b_5 = 96: q$

ANSWER 2

4. Same: b_1

ANSWER 6

5. S_6 of 3, 6, 12, ...

ANSWER 189

6. $12 + 4 + \frac{4}{3} + \dots$

ANSWER 18

7. $0.272727\dots$

ANSWER $\frac{3}{11}$

Medium · the standard question

7 PROBLEMS

1. n th term of 2, 5, 10, 17, ...

ANSWER $n^2 + 1$

2. S_{10} of 1, 4, 7, ...

ANSWER 145

3. S_5 of 2, 6, 18, ...

ANSWER 242

4. An infinite GP with $b_1 = 20$, $q = 0.2$

ANSWER 25

5. 0.454545...

ANSWER $\frac{5}{11}$

6. Which term of 4, 10, 16, ... is 100?

ANSWER The 17th

7. Which term of 3, 6, 12, ... is 384?

ANSWER The 8th

Hard · stretch and reason

7 PROBLEMS

1. Three numbers in AP: sum 24, and the squares sum to 200

ANSWER 6, 8, 10

2. Three numbers in GP: sum 14, product 64

ANSWER 2, 4, 8

3. $S_4 = 30, S_8 = 510$ in a GP: q

ANSWER 2

4. An infinite GP has $S = 16$ and $b_2 = 4$: find q

ANSWER $\frac{1}{2}$

5. n th term of 1, 5, 12, 22, ...

ANSWER $\frac{n(3n-1)}{2}$

6. A ball from 4 m rebounds $\frac{1}{2}$: total distance

ANSWER 12 m

7. Sum $1 - \frac{1}{2} + \frac{1}{4} - \dots$

ANSWER $\frac{2}{3}$

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the probability chapter begins.

1. In an AP, $a_3 = 10$ and $a_7 = 26$. Find a_1 , d and S_{10} .
2. In a GP, $b_1 = 5$ and $q = 3$. Find b_5 and S_5 .
3. Sum $18 + 6 + 2 + \dots$ to infinity.

4. Write $0.181818\dots$ as a fraction.
5. Find the n th term of $3, 9, 19, 33, \dots$

Events, and the kinds of event

Before any number can be attached to chance, the language has to be exact.

LESSON 88–89

2 × 40 MIN

ALGEBRA 9, §35

IGX 8.1

LESSON CLOCK

13'

20'

23'

20'

4'

Trial, outcome, event	13 min
Three kinds of event	20 min
Relations between events	23 min
The sample space	20 min
Homework	4 min

LEARNING OBJECTIVES

- Distinguish a trial, an outcome and an event.
- Classify events as certain, impossible or random.
- Decide whether two events are mutually exclusive, and what the opposite event is.
- List the sample space of a simple experiment.

TEXTBOOK

National	pp. 211–216
Cambridge	Core & Extended, pp. 158–163
Workbook	Exercise 8.1

01

Explanation

Trial, outcome, event

WORD	MEANS	EXAMPLE
trial	the experiment performed once	rolling a die
outcome	one indivisible result	4
sample space	the set of all outcomes	{1,2,3,4,5,6}
event	a set of outcomes	“even”, i.e. {2,4,6}

An **event** is any collection of outcomes, so it is a **subset** of the sample space — which is why the set notation of Quarter II reappears here almost unchanged.

SAY THE EVENT OUT LOUD AS A SET

“An even number” is {2,4,6}; “more than 4” is {5,6}; “even and more than 4” is {6}. Writing the set removes almost every ambiguity that word problems in this chapter contain.

Three kinds of event

KIND	HAPPENS	AS A SET	EXAMPLE (A DIE)
certain	always	the whole sample space	“less than 7”
impossible	never	$\emptyset$	“equal to 8”
random	sometimes	a proper non-empty subset	“even”

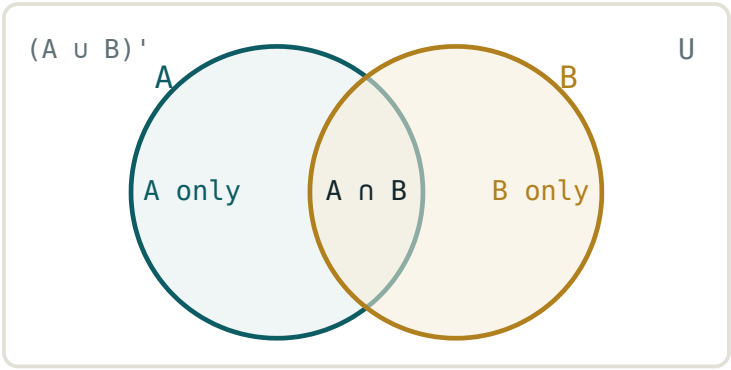
Only random events are interesting, but naming the other two matters: a probability of 1 means certain, a probability of 0 means impossible, and every random event lies strictly between.

“UNLIKELY” IS NOT “IMPOSSIBLE”

Rolling six sixes in a row is unlikely — probability $\frac{1}{46656}$ — but perfectly possible. Rolling a 7 is impossible. The two words describe different things, and the first chapter of probability exists largely to separate them.

Relations between events

RELATION	MEANS	IN SETS	EXAMPLE (A DIE)
mutually exclusive	cannot both happen	$A \cap B = \emptyset$	“even” and “equal to 3”
compatible	can both happen	$A \cap B \neq \emptyset$	“even” and “more than 3”
complementary	exactly one must happen	$B = A'$	“even” and “odd”
A implies B	A forces B	$A \subset B$	“equal to 6” and “even”



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

*Mutually exclusive events do not overlap;
complementary events fill the space between them.*

Complementary events are mutually exclusive **and** cover everything. Two events can be mutually exclusive without being complementary — “equal to 1” and “equal to 2”, for example.

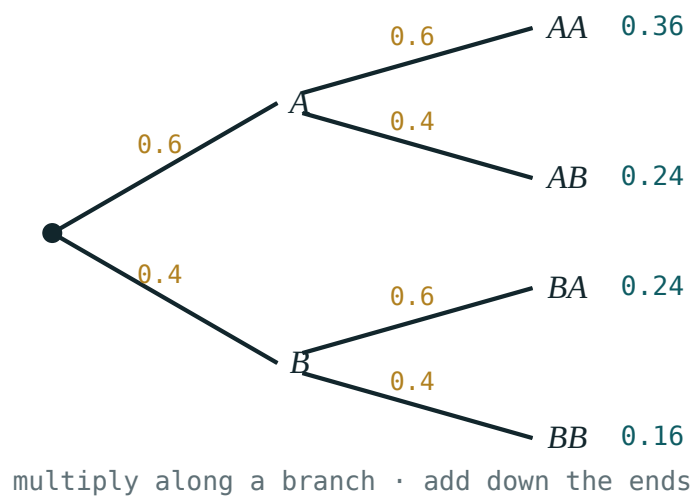
THE OPPOSITE OF “AT LEAST ONE” IS “NONE”

Not “at most one”, and not “exactly one”. Getting the complement wrong is the single most common error in probability word problems, and it is a language error, not a mathematical one.

The sample space

Listing all outcomes is the first step in nearly every probability question.

EXPERIMENT	SAMPLE SPACE	SIZE
one coin	{H, T}	2
two coins	{HH, HT, TH, TT}	4
one die	{1,...,6}	6
two dice	all ordered pairs	36
a card from a pack	all cards	52



A tree diagram lists the sample space of a two-stage experiment without missing a branch.

HT AND TH ARE DIFFERENT OUTCOMES

With two coins there are four outcomes, not three. Treating “one head and one tail” as a single outcome makes it twice as likely as the others — and this is exactly the mistake that made probability a hard subject for two centuries.

02 Worked examples

EXAMPLE 1 A die is rolled. Classify: (a) “less than 7”, (b) “equal to 8”, (c) “even”.

- ① (a) The set is $\{1, \dots, 6\}$ — everything.
Certain.

- ② (b) The set is $\emptyset$.
Impossible.

- ③ (c) The set is $\{2, 4, 6\}$.
Random.

- ④ Certain, impossible, random.

ANSWER

Certain, impossible, random

EXAMPLE 2 Are “even” and “more than 3” mutually exclusive on a die?

- 1 “even” = $\{2, 4, 6\}$.
- 2 “more than 3” = $\{4, 5, 6\}$.
- 3 $\{2, 4, 6\} \cap \{4, 5, 6\} = \{4, 6\} \neq \emptyset$
- 4 No — they are compatible.

ANSWER

No — they can both happen

EXAMPLE 3 Two coins are tossed. List the sample space and the event “at least one head”.

- 1 Sample space: $\{HH, HT, TH, TT\}$.
Four outcomes.
- 2 “at least one head” = $\{HH, HT, TH\}$.
- 3 Its opposite is “no heads” = $\{TT\}$.
- 4 Three outcomes out of four.

ANSWER $\{HH, HT, TH\}$ — 3 of 4**03 Interactive model**

Toss two coins twenty times and tally HH, HT, TH, TT; the class sees that “one of each” occurs about half the time, and the four-outcome sample space explains why.

Certain, impossible, or random?

Throwing a die and getting a number under 7 is:

Getting a 7 on one die is:

Getting an even number is:

The probability of a certain event is:

0

0.5

1

undefined

The probability of an impossible event is:

0

0.5

1

-1

Two events that cannot both happen are:

independent

mutually exclusive

complementary always

certain

An event and its complement have probabilities that:

are equal

add to 1

multiply to 1

add to 0

Every probability lies:

between 0 and 1

above 1

below 0

anywhere

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Trial	Sinov	Испытание
Outcome	Natija	Исход
Event	Hodisa	Событие
Certain event	Muqarrar hodisa	Достоверное событие
Impossible event	Mumkin bo'lmagan hodisa	Невозможное событие
Random event	Tasodifiy hodisa	Случайное событие
Mutually exclusive	Birgalikda bo'lmagan	Несовместные
Complementary event	Qarama-qarshi hodisa	Противоположное событие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An event is:

one outcome

a set of outcomes

the whole trial

a number

“Rolling a 7 on a die” is:

certain

impossible

random

unlikely

Mutually exclusive means:

$$A \cap B = \emptyset$$

$$A \cup B = \xi$$

$$A \subset B$$

$$A = B$$

The opposite of “at least one head” is:

at most one head

exactly one head

no heads

two heads

Two coins have how many outcomes?

2

3

4

6

A certain event has probability:

0

$\frac{1}{2}$

1

undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. “less than 7” on a die

ANSWER Certain

2. “equal to 8” on a die

ANSWER Impossible

3. “even” on a die

ANSWER Random

4. Sample space of one coin

ANSWER $\{H, T\}$

5. Sample space of one die

ANSWER $\{1, \dots, 6\}$

6. Number of outcomes for two coins

ANSWER 4

7. Number of outcomes for two dice

ANSWER 36

Medium · the standard question

7 PROBLEMS

1. “even” as a set

ANSWER $\{2,4,6\}$

2. “more than 3” as a set

ANSWER $\{4,5,6\}$

3. Are they mutually exclusive?

ANSWER No

4. Opposite of “even”

ANSWER “odd”

5. Opposite of “at least one head”

ANSWER “no heads”

6. “at least one head” with two coins

ANSWER $\{HH, HT, TH\}$

7. Is “equal to 6” $\subset$ “even”?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. Two dice: outcomes with a sum of 7

ANSWER 6

2. Two dice: outcomes with a sum of 2

ANSWER 1

3. Opposite of “at most two heads” in three tosses

ANSWER “three heads”

4. Three coins: how many outcomes?

ANSWER 8

5. Three coins: “exactly two heads” as a set

ANSWER $\{HHT, HTH, THH\}$

6. Give two mutually exclusive but not complementary events on a die

ANSWER “= 1” and “= 2”

7. A card: are “red” and “king” mutually exclusive?

ANSWER No — there are two red kings

07 Homework

Homework — 5 tasks

Write every event as a set before answering anything about it.

1. Classify on a die: “more than 0”, “equal to 0”, “odd”.
2. List the sample space for three coins.
3. Write “at least two heads” in three tosses as a set.
4. Give the opposite of “at least one six” in two rolls.
5. Give an example of two compatible events on a die, and two mutually exclusive ones.

The classical definition of probability

Count the favourable outcomes, count them all, and divide — but only when they are equally likely.

LESSON 90–91

2 × 40 MIN

ALGEBRA 9, §36

IGX 8.2

LESSON CLOCK

13'

20'

23'

20'

4'

The definition	13 min
What the number means	20 min
Counting carefully	23 min
The complement	20 min
Homework	4 min

LEARNING OBJECTIVES

- State and apply $P(A) = m/n$, naming the condition of equal likelihood.
- Know that $0 \leq P(A) \leq 1$ and that $P(A) + P(A') = 1$.
- Compute probabilities for dice, coins, cards and coloured balls.
- Use the complement when it is quicker than direct counting.

TEXTBOOK

National	pp. 217–223
Cambridge	Core & Extended, pp. 164–170
Workbook	Exercise 8.2

01

Explanation

The definition

If a trial has n **equally likely** outcomes and m of them belong to the event A , then

$$P(A) = \frac{m}{n} \text{ — } \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

EVENT	M	N	P
a die shows 4	1	6	$\frac{1}{6}$
a die shows an even number	3	6	$\frac{1}{2}$
two coins give two heads	1	4	$\frac{1}{4}$
a card is a heart	13	52	$\frac{1}{4}$

“EQUALLY LIKELY” IS A CONDITION, NOT A DECORATION

A drawing pin lands point-up or point-down: two outcomes, but not equally likely, so the answer is **not** $\frac{1}{2}$. The classical definition simply does not apply, and the next lesson supplies what does.

What the number means

PROPERTY	STATEMENT	WHY
bounds	$0 \leq P(A) \leq 1$	$0 \leq m \leq n$
impossible	$P(\emptyset) = 0$	$m = 0$
certain	$P(\xi) = 1$	$m = n$
complement	$P(A) + P(A') = 1$	every outcome is in one or the other

A probability is never negative, never greater than 1, and is best given as a fraction in lowest terms unless the question asks for a decimal or a percentage.

AN ANSWER OF $\frac{7}{6}$ OR -0.2 IS NOT A SMALL SLIP

It is a signal that the counting went wrong — usually that n was taken too small. Check every probability against $0 \leq P \leq 1$ before writing it down.

Counting carefully

Almost every mark in this topic is a counting mark.

EXPERIMENT	N	A TRAP
two dice	36	$(2,5)$ and $(5,2)$ are different
two coins	4	$HT \neq TH$
drawing 2 balls of 10, without replacement	90	n changes after the first draw
a card from a pack	52	four suits, thirteen ranks

Example. Two dice. $P(\text{sum} = 7)$: the favourable pairs are $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ — six of them — so $P = \frac{6}{36} = \frac{1}{6}$. A sum of 7 is the most likely total, and this counting is why.

DRAW THE 6×6 GRID ONCE

Every two-dice question in the Grade 9 course can be read off a single 6×6 table of sums. Drawing it once, at the start of the exercise, turns ten questions into ten readings.

The complement

When an event contains many outcomes and its opposite contains few, count the opposite.

$P(A) = 1 - P(A')$

EVENT	DIRECT COUNT	BY COMPLEMENT
at least one six in two rolls	11	$1 - \frac{25}{36} = \frac{11}{36}$
at least one head in three tosses	7	$1 - \frac{1}{8} = \frac{7}{8}$
a card that is not a heart	39	$1 - \frac{1}{4} = \frac{3}{4}$

“AT LEAST ONE” ALMOST ALWAYS MEANS “ONE MINUS NONE”

Counting the ways to get at least one six means adding several cases; counting the ways to get none means one multiplication. Recognising the phrase is worth more marks in this chapter than any formula.

02 Worked examples

EXAMPLE 1 A die is rolled. Find $P(\text{even})$ and $P(\text{more than 4})$.

- ① $n = 6$ equally likely outcomes.
- ② $\text{even} = \{2, 4, 6\}$, so $m = 3$.
- ③ $P = \frac{3}{6} = \frac{1}{2}$
- ④ $\text{more than 4} = \{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

ANSWER

$$\frac{1}{2} \text{ and } \frac{1}{3}$$

EXAMPLE 2 Two dice are rolled. Find $P(\text{the sum is 7})$.

- ① $n = 6 \times 6 = 36$
Ordered pairs.
- ② Favourable: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$.
- ③ $m = 6$
- ④ $P = \frac{6}{36} = \frac{1}{6}$

ANSWER

$$\frac{1}{6}$$

EXAMPLE 3 Two dice are rolled. Find $P(\text{at least one six})$.

① The opposite is “no six at all”.

② No six: $5 \times 5 = 25$ outcomes.

③ $P(\text{no six}) = \frac{25}{36}$.

④ $P = 1 - \frac{25}{36} = \frac{11}{36}$

ANSWER

$$\frac{11}{36}$$

03 Interactive model

Draw the 6×6 grid of dice sums on the board and leave it there for the whole chapter; almost every question becomes a matter of counting cells.

Count, count, divide

One die: $P(\text{an even number})$ is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

One die: $P(\text{a number over } 4)$ is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

A pack of 52 cards: $P(\text{a heart})$ is:

$$\frac{1}{13}$$

$$\frac{1}{4}$$

$$\frac{1}{2}$$

$$\frac{1}{52}$$

Two coins: $P(\text{two heads})$ is:

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{1}{4}$$

$$\frac{3}{4}$$

Two dice: how many outcomes in total?

12

24

36

6

Two dice: $P(\text{a total of } 7)$ is:

$$\frac{1}{6}$$

$$\frac{1}{9}$$

$$\frac{1}{12}$$

$$\frac{5}{36}$$

A bag of 4 red and 6 blue: $P(\text{red})$ is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{4}{6}$$

$$\frac{1}{4}$$

If $P(A) = 0.3$ then $P(\text{not } A)$ is:

$$0.3$$

$$0.7$$

$$1.3$$

$$0$$

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Probability	Ehtimollik	Вероятность
Favourable outcome	Qulay natija	Благоприятный исход
Equally likely	Teng imkoniyatli	Равновозможные
Classical definition	Klassik ta'rif	Классическое определение
Complement rule	Qarama-qarshi hodisa qoidasi	Правило дополнения
At least	Kamida	Хотя бы
At most	Ko'pi bilan	Не более
Without replacement	Qaytarmasdan	Без возвращения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$P(A)$ equals:

$$\frac{n}{m}$$

$$\frac{m}{n}$$

$$m n$$

$$m - n$$

The definition requires the outcomes to be:

few

equally likely

independent

positive

P of a certain event:

$$0$$

$$\frac{1}{2}$$

$$1$$

$$\infty$$

$P(A) + P(A')$ equals:

0

$\frac{1}{2}$

1

2

Two dice: $P(\text{sum } 7) =$

$\frac{1}{6}$

$\frac{1}{9}$

$\frac{1}{12}$

$\frac{1}{36}$

“At least one” is best done by:

direct counting

the complement

a tree

guessing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $P(\text{a die shows 4})$

ANSWER $\frac{1}{6}$

2. $P(\text{a die shows even})$

ANSWER $\frac{1}{2}$

3. $P(\text{a die shows more than 4})$

ANSWER $\frac{1}{3}$

4. $P(\text{a coin shows heads})$

ANSWER $\frac{1}{2}$

5. $P(\text{two coins give two heads})$

ANSWER $\frac{1}{4}$

6. $P(\text{a card is a heart})$

ANSWER $\frac{1}{4}$

7. $P(\text{a die shows 7})$

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Two dice: $P(\text{sum } 7)$

ANSWER $\frac{1}{6}$

2. Two dice: $P(\text{sum } 2)$

ANSWER $\frac{1}{36}$

3. Two dice: $P(\text{at least one six})$

ANSWER $\frac{11}{36}$

4. Three coins: $P(\text{exactly two heads})$

ANSWER $\frac{3}{8}$

5. Three coins: $P(\text{at least one head})$

ANSWER $\frac{7}{8}$

6. A bag: 4 red, 6 blue. $P(\text{red})$

ANSWER $\frac{2}{5}$

7. A card: $P(\text{a king})$

ANSWER $\frac{1}{13}$

Hard · stretch and reason

7 PROBLEMS

1. Two dice: $P(\text{sum is even})$

ANSWER $\frac{1}{2}$

2. Two dice: $P(\text{the numbers differ by 1})$

ANSWER $\frac{5}{18}$

3. A bag: 3 red, 5 blue; two drawn without replacement. $P(\text{both red})$

ANSWER $\frac{3}{28}$

4. Same bag: $P(\text{one of each})$

ANSWER $\frac{15}{28}$

5. A card: $P(\text{red or a king})$

ANSWER $\frac{7}{13}$

6. Three dice: $P(\text{at least one six})$

ANSWER $\frac{91}{216}$

7. A number from 1 to 20: $P(\text{a multiple of 3 or of 5})$

ANSWER $\frac{9}{20}$

07 Homework

Homework — 5 tasks

Write n and m explicitly before dividing, and check that $0 \leq P \leq 1$.

1. A die is rolled. Find $P(\text{odd})$, $P(\text{less than 3})$ and $P(\text{a multiple of 3})$.
2. Two dice are rolled. Find $P(\text{sum 8})$ and $P(\text{a double})$.

3. Three coins are tossed. Find $P(\text{exactly one head})$.
4. A bag holds 5 red and 7 green balls. Find $P(\text{green})$.
5. Find $P(\text{at least one head in four tosses})$ using the complement.

Relative frequency and statistical probability

When outcomes are not equally likely, count what actually happens instead.

LESSON 92–93

2 × 40 MIN

ALGEBRA 9, §37

IGX 8.3

LESSON CLOCK

13'

20'

23'

20'

4'

When counting outcomes fails	13 min
Relative frequency	20 min
Why it settles	23 min
Estimating with it	20 min
Homework	4 min

LEARNING OBJECTIVES

- Define relative frequency and compute it from a table of results.
- Explain why relative frequency settles down as the number of trials grows.
- Use relative frequency to estimate a probability, and to estimate expected numbers.
- Say when the classical definition applies and when it does not.

TEXTBOOK

National	pp. 224–229
Cambridge	Core & Extended, pp. 171–176
Workbook	Exercise 8.3

01

Explanation

When counting outcomes fails

The classical definition needs equally likely outcomes. Very often they are not.

TRIAL	OUTCOMES	EQUALLY LIKELY?
a fair die	6	yes
a drawing pin	2	no
a weighted die	6	no
whether it rains tomorrow	2	no
whether a bulb lasts 1000 hours	2	no

For all but the first, the probability cannot be calculated in advance. It must be **measured**.

TWO DEFINITIONS, TWO SITUATIONS

The classical definition is used when the symmetry of the object tells you the answer; the statistical one when only experiment can. Neither replaces the other, and saying which one a question needs is itself a mark.

Relative frequency

Perform the trial N times and let the event happen M times. The **relative frequency** is

$$W(A) = \frac{M}{N}$$

Example. A drawing pin is dropped 200 times and lands point-up 124 times. Then $W = \frac{124}{200} = 0.62$, and 0.62 is the best available estimate of the probability.

RESULT	FREQUENCY	RELATIVE FREQUENCY
point up	124	0.62
point down	76	0.38
total	200	1.00

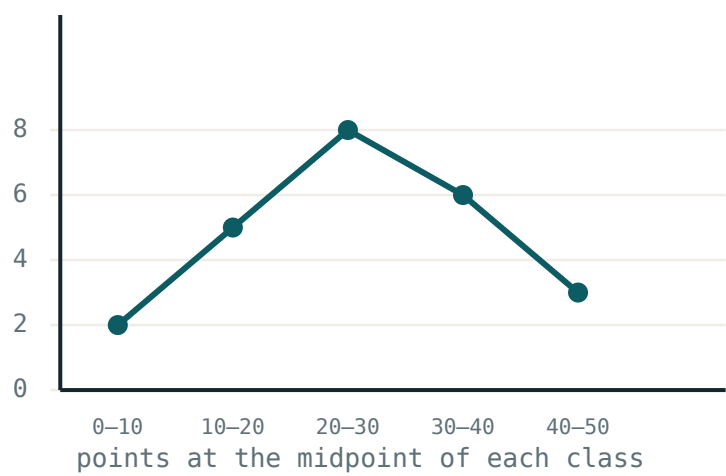
THE RELATIVE FREQUENCIES OF ALL OUTCOMES ADD TO 1

Exactly as probabilities do. A table whose last column does not total 1 contains an arithmetic error, and checking it takes three seconds.

Why it settles

Repeat the experiment in blocks and watch the relative frequency:

N	10	50	100	500	5000
W	0.70	0.58	0.63	0.615	0.618



As the number of trials grows the relative frequency wanders less and less.

The values jump about at first and then settle. This **stability of relative frequency** is an experimental fact, observed for three centuries, and it is what makes probability a science rather than an opinion.

TEN TRIALS PROVE NOTHING

Getting seven heads in ten tosses is entirely ordinary for a fair coin. Declaring the coin biased on that evidence is the classic error; a few hundred trials are needed before a departure from 0.5 means anything.

Estimating with it

Once P is estimated, the expected number of occurrences in N future trials is $N \times P$.

QUESTION	WORKING	ANSWER
the pin, 500 more drops	500×0.62	310
a die, $P(\text{six}) = \frac{1}{6}$, 300 rolls	$300 \times \frac{1}{6}$	50
3% of bulbs are faulty, 2000 bulbs	2000×0.03	60

“EXPECTED” MEANS ON AVERAGE, NOT EXACTLY

Fifty sixes in 300 rolls is what the arithmetic predicts; 47 or 55 would surprise nobody. The expected number is the centre of a spread, and a question asking “how many would you expect” is asking for that centre.

02 Worked examples

EXAMPLE 1 A pin is dropped 200 times and lands point-up 124 times. Estimate the probability, and predict the point-up count in 500 more drops.

$$1 \quad W = \frac{124}{200}$$

$$2 \quad = 0.62$$

The estimate.

$$3 \quad 500 \times 0.62$$

$$4 \quad = 310 \text{ times.}$$

Expected, not exact.

ANSWER

0.62; about 310

EXAMPLE 2 A die is rolled 600 times; a six appears 92 times. Is there evidence of bias?

$$1 \quad W = \frac{92}{600} \approx 0.153$$

$$2 \quad \text{A fair die would give } \frac{1}{6} \approx 0.167.$$

$$3 \quad \text{Expected count } 100; \text{ observed } 92.$$

$$4 \quad \text{The difference is small for } 600 \text{ trials — no clear evidence.}$$

More trials would be needed.

ANSWER

No clear evidence of bias

EXAMPLE
3**In a school of 800, a survey of 50 pupils found 18 walk to school. Estimate how many walk in total.**

$$① \quad w = \frac{18}{50} = 0.36$$

$$② \quad 800 \times 0.36$$

$$③ \quad = 288$$

$$④ \quad \text{About 288 pupils.}$$

A sample of 50 is small — treat it as an estimate.

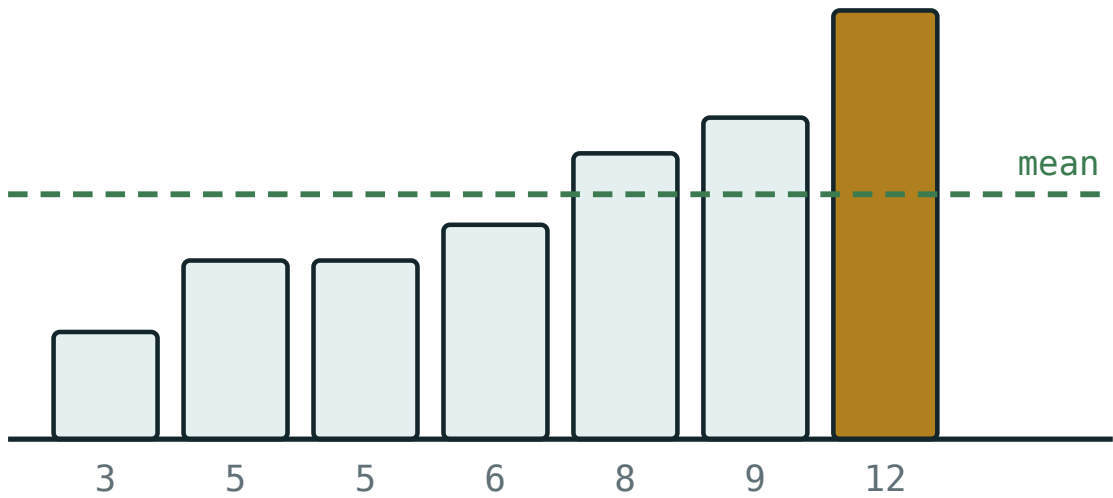
ANSWER

About 288

03 Interactive model

Have every pupil toss a coin twenty times and pool the results on the board; the class watches the pooled relative frequency approach 0.5 while individual results scatter wildly.

Relative frequency as trials grow



values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Relative frequency	Nisbiy chastota	Относительная частота
Frequency	Chastota	Частота
Number of trials	Sinovlar soni	Число испытаний
Estimate	Baholash	Оценка
Stabilises	Barqarorlashadi	Стабилизируется
Expected number	Kutilayotgan son	Ожидаемое число
Biased	Nosimmetrik	Смещённый
Fair	Simmetrik	Правильный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Relative frequency is:

$$\frac{M}{N}$$

$$\frac{N}{M}$$

$$M N$$

$$M - N$$

It is used when outcomes are:

equally likely

not equally likely

few

many

As N grows, W :

grows

settles

falls

oscillates forever

The relative frequencies of all outcomes total:

0

$\frac{1}{2}$

1

N

Expected number in N trials:

$N + P$

$N \times P$

$\frac{N}{P}$

P

Seven heads in ten tosses shows:

bias

nothing much

a fair coin

an error

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $M = 124, N = 200: W$

ANSWER 0.62

2. $M = 30, N = 50: W$

ANSWER 0.6

3. $M = 9, N = 20: W$

ANSWER 0.45

4. If $W(\text{up}) = 0.62, W(\text{down})$

ANSWER 0.38

5. Expected sixes in 300 rolls

ANSWER 50

6. Expected heads in 80 tosses

ANSWER 40

7. 3% faulty of 2000

ANSWER 60

Medium · the standard question

7 PROBLEMS

1. Pin: 0.62, 500 more drops

ANSWER 310

2. Survey 18 of 50 walk; school of 800

ANSWER 288

3. A die: 92 sixes in 600: W

ANSWER ≈ 0.153

4. Is that clear evidence of bias?

ANSWER No

5. A seed germinates 171 times in 200: W

ANSWER 0.855

6. Expected germinations from 1000 seeds

ANSWER 855

7. Which definition for “will it rain tomorrow”?

ANSWER Statistical

Hard · stretch and reason

7 PROBLEMS

1. A biased coin: $W(\text{head}) = 0.6$. Expected heads in 250

ANSWER 150

2. A machine: 4 faults in 500. Expected faults in 12000

ANSWER 96

3. A survey of 40 found 14 left-handed. Estimate for 1200

ANSWER 420

4. Two dice rolled 360 times: expected number of sevens

ANSWER 60

5. A spinner: W values 0.2, 0.35, x , 0.15. Find x

ANSWER 0.3

6. Why is $N = 10$ too few to judge a coin?

ANSWER The scatter is far larger than the effect

7. A die gives 40 sixes in 120. Comment.

ANSWER $W = \frac{1}{3}$ — strong evidence of bias

07 Homework

Homework — 5 tasks

Say for every task whether the classical or the statistical definition applies, and why.

1. A pin lands point-up 87 times in 150 drops. Find W and predict the count in 600 drops.
2. A survey of 60 pupils found 21 cycle to school. Estimate the number in a school of 900.
3. A die is rolled 300 times and gives 72 sixes. Comment on whether it is fair.
4. A factory finds 6 faulty items in 400. How many would you expect in 10 000?

5. Explain why the classical definition cannot be used for tomorrow's weather.

Counting for probability — the two rules and arrangements

Before a probability can be found, the outcomes must be counted — and counting has rules.

LESSON 94–95

2 × 40 MIN

ALGEBRA 9, §38

IGX 8.4

LESSON CLOCK

13'

20'

23'

20'

4'

The two rules	13 min
Arrangements	20 min
Selections	23 min
Counting into probability	20 min
Homework	4 min

LEARNING OBJECTIVES

- Use the multiplication rule for a sequence of independent choices.
- Use the addition rule for alternatives that cannot both happen.
- Count arrangements of n distinct objects and selections of k from n .
- Apply the counting to a probability question.

TEXTBOOK

National	pp. 230–236
Cambridge	Core & Extended, pp. 177–182
Workbook	Exercise 8.4

01

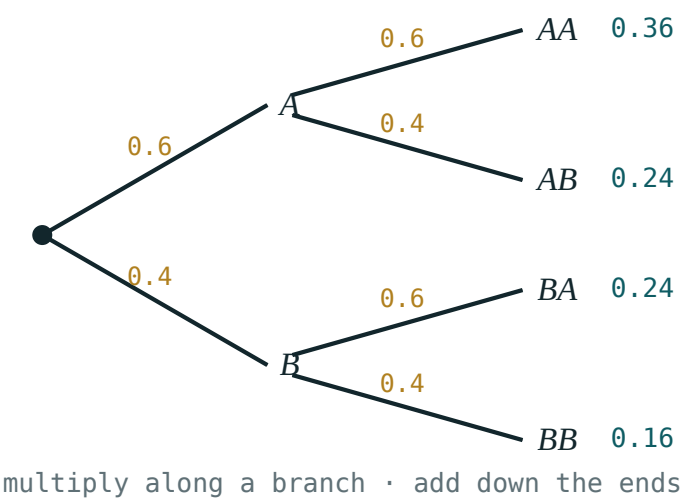
Explanation

The two rules

RULE	USE WHEN	COUNT	WORD
multiplication	choices are made one after another	$m \times n$	and
addition	the cases are alternatives	$m + n$	or

Multiplication. Four shirts and three trousers give $4 \times 3 = 12$ outfits: a shirt **and** a pair of trousers.

Addition. Travelling by one of 3 buses **or** one of 2 trains gives $3 + 2 = 5$ ways.



Each stage of a tree multiplies the branches; the leaves are all the outcomes.

FIND THE WORD “AND” OR THE WORD “OR”

Almost every counting question is settled by which of the two words describes the situation. Read the sentence aloud and listen for it before writing anything.

Arrangements

Arranging n distinct objects in a row: n choices for the first place, $n - 1$ for the second, and so on.

$n! = n(n - 1)(n - 2) \dots 2 \cdot 1$

N	1	2	3	4	5	6
$n!$	1	2	6	24	120	720

Arranging k of n in order — filling k places from n objects — gives $n(n - 1) \dots (n - k + 1)$. For example, choosing a president, a secretary and a treasurer from 10 people: $10 \times 9 \times 8 = 720$ ways.

CHOOSING WITH REPLACEMENT IS DIFFERENT

A three-digit code from 10 digits, repeats allowed, is $10^3 = 1000$, not $10 \times 9 \times 8$. Decide first whether an object can be used twice; the two answers differ by a lot.

Selections

When the order does **not** matter, every selection has been counted $k!$ times, so divide.

$$C = \frac{n(n-1) \dots (n-k+1)}{k!}$$

QUESTION	ORDER?	COUNT
3 offices from 10 people	yes	$10 \times 9 \times 8 = 720$
a committee of 3 from 10	no	$\frac{720}{6} = 120$
2 balls from 8	no	$\frac{8 \times 7}{2} = 28$
a handshake among 6 people	no	$\frac{6 \times 5}{2} = 15$

ONE QUESTION DECIDES EVERYTHING: DOES ORDER MATTER?

A committee is a set — Ali, Bek, Dilnoza is the same committee as Dilnoza, Bek, Ali. A podium is a list — first, second and third are different places. Ask the question in words before choosing a formula.

Counting into probability

With n and m both counted properly, the classical definition finishes the job.

Example. Two balls are drawn from a bag of 3 red and 5 blue. Find $P(\text{both red})$.

STEP	WORKING
all selections of 2 from 8	$\frac{8 \times 7}{2} = 28$
selections of 2 from the 3 red	$\frac{3 \times 2}{2} = 3$
probability	$\frac{3}{28}$

The same answer comes from multiplying stage by stage: $\frac{3}{8} \times \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$.

Both routes are correct, and agreeing with yourself by two methods is a good check.

COUNT N AND M THE SAME WAY

If the total is counted as ordered pairs, the favourable cases must be too. Mixing an ordered total with an unordered favourable count is the classic source of answers that are out by a factor of 2.

02

Worked examples

EXAMPLE 1 How many three-digit numbers can be made from 1, 2, 3, 4, 5 (a) with repeats, (b) without?

① (a) Each of the three places has 5 choices.

② $5^3 = 125$

③ (b) $5 \times 4 \times 3$.

No digit reused.

④ $= 60$

ANSWER

125 and 60

EXAMPLE 2 A committee of 3 is chosen from 10 people. In how many ways?

- ① Order does not matter.
- ② Ordered count: $10 \times 9 \times 8 = 720$.
- ③ Each committee counted $3! = 6$ times.
- ④ $\frac{720}{6} = 120$

ANSWER

120

EXAMPLE 3 Two balls are drawn from 3 red and 5 blue. Find $P(\text{both red})$.

- ① All selections: $\frac{8 \times 7}{2} = 28$.
- ② Red selections: $\frac{3 \times 2}{2} = 3$.
- ③ $P = \frac{3}{28}$
- ④ Check: $\frac{3}{8} \times \frac{2}{7} = \frac{3}{28} \checkmark$
Two routes agree.

ANSWER

$\frac{3}{28}$

03 Interactive model

Ask how many ways three pupils can stand in a row, then how many ways three can be chosen for a team; the class discovers the division by $3!$ themselves.

And, or, and does order matter?

A choice of 3 shirts **and** 4 ties gives:

A choice of 3 shirts **or** 4 ties gives:

Arrangements of 4 different books on a shelf:

$5!$ equals:

25

60

120

720

Three-digit codes from 10 digits, repeats allowed:

30

720

1000

120

Three-digit codes with no repeats:

720

1000

120

30

When order does not matter you are counting:

arrangements

selections

outcomes

events

Choosing 2 pupils from 5, order not counted:

10

20

25

5

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Addition rule	Qo'shish qoidasi	Правило сложения
Arrangement	O'rin almashtirish	Перестановка
Selection	Tanlash	Выбор
Factorial	Faktorial	Факториал
Order matters	Tartib muhim	Порядок важен
With replacement	Qaytarib	С возвращением
Tree diagram	Daraxt diagrammasi	Дерево вариантов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“And” means:

add

multiply

divide

subtract

“Or” (exclusive alternatives) means:

add

multiply

divide

subtract

$4!$ equals:

12

16

24

64

A committee of 3 from 10:

720

120

30

1000

Three offices from 10 people:

720

120

30

1000

A three-digit code from 10 digits, repeats allowed:

720

120

1000

30

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 4 shirts, 3 trousers: outfits

ANSWER 12

2. 3 buses or 2 trains: ways

ANSWER 5

3. $3!$

ANSWER 6

4. $4!$

ANSWER 24

5. $5!$

ANSWER 120

6. Arrangements of 3 pupils in a row

ANSWER 6

7. Two coins: outcomes

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Three-digit numbers from 1..5, repeats allowed

ANSWER 125

2. Same, no repeats

ANSWER 60

3. Three offices from 10 people

ANSWER 720

4. A committee of 3 from 10

ANSWER 120

5. Handshakes among 6 people

ANSWER 15

6. Selections of 2 from 8

ANSWER 28

7. $P(\text{both red})$ from 3 red, 5 blue

ANSWER $\frac{3}{28}$

Hard · stretch and reason

7 PROBLEMS

1. Arrangements of the letters of *KITOB*

ANSWER 120

2. Four-digit codes from 0–9, repeats allowed

ANSWER 10000

3. A committee of 2 boys and 2 girls from 5 and 4

ANSWER 60

4. P (a committee of 3 from 5 boys and 5 girls is all boys)

ANSWER $\frac{1}{12}$

5. Diagonals of a decagon

ANSWER 35

6. Two cards from a pack: P (both aces)

ANSWER $\frac{1}{221}$

7. How many ways can 5 pupils sit if two insist on sitting together?

ANSWER 48

07 Homework

Homework — 5 tasks

Write “order matters” or “order does not matter” at the top of every solution.

- How many four-digit numbers can be made from 1, 2, 3, 4, 5, 6 without repeats?
- In how many ways can a committee of 4 be chosen from 9 people?
- How many handshakes take place among 8 people?
- Two balls are drawn from 4 red and 6 blue. Find P (both blue).

5. How many arrangements has the word *DAFTAR* if all letters are treated as distinct?

Chapter exercises — Cambridge probability with tree diagrams

Two-stage experiments, with and without replacement — the IGCSE way of laying them out.

LESSON 96–97

2 × 40 MIN

ALGEBRA 9, V BOB MASHQLARI

IGX 8.5–8.6

LESSON CLOCK

13'

20'

23'

20'

4'

Drawing the tree	13 min
Along and between	20 min
Without replacement	23 min
Tables and Venn diagrams	20 min
Homework	4 min

LEARNING OBJECTIVES

- Draw a tree diagram for a two-stage experiment and label every branch.
- Multiply along a path and add between paths.
- Adjust the second-stage probabilities when there is no replacement.
- Use a Venn diagram or a two-way table where it is quicker than a tree.

TEXTBOOK

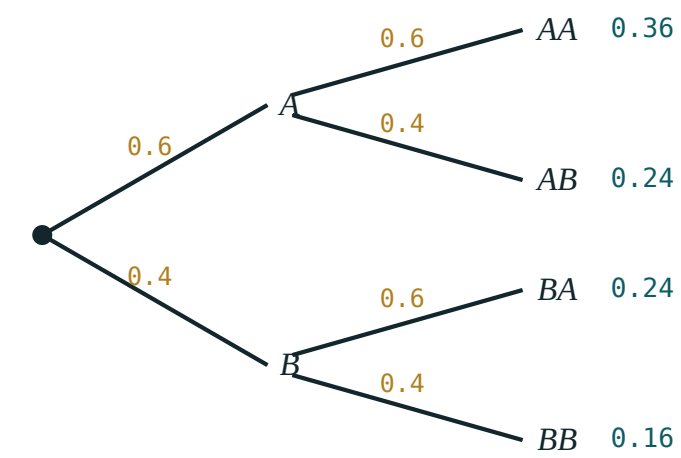
National	pp. 237–240
Cambridge	Core & Extended, pp. 183–192
Workbook	Exercise 8.5–8.6

01

Explanation

Drawing the tree

A tree diagram sets out a two-stage experiment so that no case is missed.



multiply along a branch · add down the ends

Two stages, four paths — the probabilities on each set of branches add to 1.

RULE	CHECK
each set of branches	its probabilities add to 1
along a path	multiply
between paths	add
all the paths together	add to 1

LABEL THE BRANCHES BEFORE COMPUTING ANYTHING

Cambridge awards marks for a correctly labelled tree even when the arithmetic afterwards goes wrong. Drawing it is never wasted time.

Along and between

Example. A bag holds 3 red and 5 blue balls. One is drawn, replaced, and another drawn.

PATH	PROBABILITY
RR	$\frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$
RB	$\frac{3}{8} \times \frac{5}{8} = \frac{15}{64}$
BR	$\frac{5}{8} \times \frac{3}{8} = \frac{15}{64}$
BB	$\frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$
total	$\frac{64}{64} = 1 \checkmark$

Then $P(\text{one of each}) = \frac{15}{64} + \frac{15}{64} = \frac{30}{64} = \frac{15}{32}$, and $P(\text{at least one red}) = 1 - \frac{25}{64} = \frac{39}{64}$.

TWO PATHS FOR “ONE OF EACH”, NOT ONE

RB and BR are different paths and both must be counted. Forgetting the second is the commonest tree-diagram error, and the total-equals-1 check catches it every time.

Without replacement

If the first ball is not replaced, the second stage has one fewer ball — and one fewer of whichever colour was drawn.

PATH	WITH REPLACEMENT	WITHOUT REPLACEMENT
RR	$\frac{3}{8} \times \frac{3}{8}$	$\frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$
RB	$\frac{3}{8} \times \frac{5}{8}$	$\frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$
BR	$\frac{5}{8} \times \frac{3}{8}$	$\frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$
BB	$\frac{5}{8} \times \frac{5}{8}$	$\frac{5}{8} \times \frac{4}{7} = \frac{5}{14}$

The denominators of the second stage are all 7; the numerators depend on the first branch. That is the whole difference, and it is where the marks are.

ONLY THE SECOND-STAGE BRANCHES CHANGE

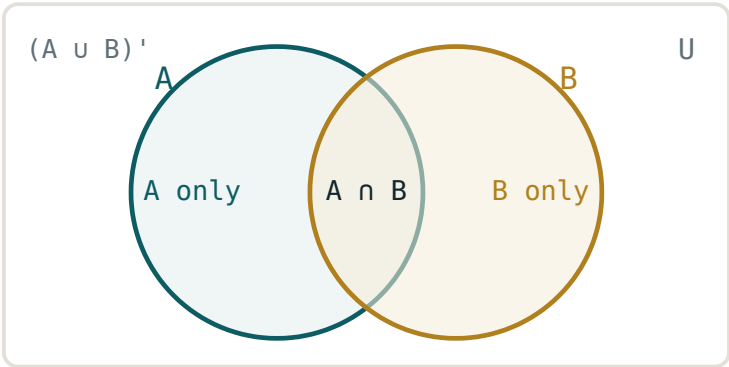
The first stage is untouched by the question of replacement — nothing has been removed yet. And the four paths must still total 1: $\frac{3}{28} + \frac{15}{56} + \frac{15}{56} + \frac{5}{14} = 1 \checkmark$

Tables and Venn diagrams

Two other layouts appear constantly in IGCSE papers.

	BOYS	GIRLS	TOTAL
Walk	12	9	21
Bus	8	11	19
Total	20	20	40

From the table, $P(\text{a girl who walks}) = \frac{9}{40}$ and $P(\text{walks}) = \frac{21}{40}$.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

A Venn diagram answers “both”, “either” and “neither” at a glance.

CHOOSE THE LAYOUT THAT FITS THE QUESTION

A tree for stages, a two-way table for two classifications of the same people, a Venn diagram for overlapping sets. Using the wrong one is not wrong, only slow — and slowness costs marks in a timed paper.

02 Worked examples

EXAMPLE 1 A bag holds 3 red and 5 blue. Two are drawn with replacement. Find $P(\text{one of each})$.

① Paths RB and BR.

Two paths, not one.

② $RB = \frac{3}{8} \times \frac{5}{8} = \frac{15}{64}$

③ $BR = \frac{5}{8} \times \frac{3}{8} = \frac{15}{64}$

④ $P = \frac{30}{64} = \frac{15}{32}$

ANSWER

$$\frac{15}{32}$$

EXAMPLE 2 The same bag, two drawn without replacement. Find $P(\text{both blue})$.

① First blue: $\frac{5}{8}$.

② Then 4 blue of 7 left: $\frac{4}{7}$.

③ $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$

④ $= \frac{5}{14}$

ANSWER

$$\frac{5}{14}$$

EXAMPLE 3 From the two-way table above, find $P(\text{a boy, given the pupil walks})$.

① Restrict to the “Walk” row: 21 pupils.
A conditional question.

② Of those, 12 are boys.

③ $P = \frac{12}{21}$

④ $= \frac{4}{7}$

ANSWER

$$\frac{4}{7}$$

03 **Interactive model**

Draw the same experiment twice, with and without replacement, on one board; the only cells that change are the second-stage branches, and seeing that is the lesson.

Trees, tables and Venn diagrams

Along a branch of a tree you:

Between separate branches you:

The probabilities on the branches from one point add to:

Two coins: $P(\text{exactly one head})$ is:

$$\frac{1}{4}$$

$$\frac{1}{2}$$

$$\frac{3}{4}$$

$$1$$

$P(\text{red then red})$ from a bag of 3 red and 2 blue, with replacement:

$$\frac{9}{25}$$

$$\frac{3}{10}$$

$$\frac{6}{25}$$

$$\frac{3}{5}$$

Without replacement it becomes:

$$\frac{9}{25}$$

$$\frac{3}{10}$$

$$\frac{6}{20}$$

$$\frac{1}{2}$$

Independent events are those where:

one changes the other

neither changes the other

both are certain

both are impossible

$P(\text{at least one head})$ in two throws is easiest as:

$1 - P(\text{no heads})$

$P(\text{one head})$

$2 \cdot P(\text{head})$

$P(\text{two heads})$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Tree diagram	Daraxt diagrammasi	Дерево вероятностей
Branch	Shox	Ветвь
Path	Yo'l	Путь
Independent events	Bog'liqsiz hodisalar	Независимые события
Dependent events	Bog'liq hodisalar	Зависимые события
With replacement	Qaytarib	С возвращением
Without replacement	Qaytarmasdan	Без возвращения
Two-way table	Ikki yo'nalishli jadval	Таблица сопряжённости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Along a path you:

add

multiply

divide

subtract

Between paths you:

add

multiply

divide

subtract

All the paths together total:

0

$\frac{1}{2}$

1

it varies

Without replacement, what changes?

the first stage

the second stage

both

neither

“One of each” uses:

one path

two paths

three paths

all paths

For two classifications of the same people, use:

a tree

a two-way table

a bar chart

a formula

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Along a path you

ANSWER Multiply

2. Between paths you

ANSWER Add

3. Branches from one point total

ANSWER 1

4. $P(RR)$ with replacement, 3 red of 8

ANSWER $\frac{9}{64}$

5. $P(BB)$ with replacement, 5 blue of 8

ANSWER $\frac{25}{64}$

6. $P(\text{two heads in two tosses})$

ANSWER $\frac{1}{4}$

7. $P(\text{a girl who walks})$ from the table

ANSWER $\frac{9}{40}$

Medium · the standard question

7 PROBLEMS

1. $P(\text{one of each}), \text{ with replacement}$

ANSWER $\frac{15}{32}$

2. $P(\text{at least one red}), \text{ with replacement}$

ANSWER $\frac{39}{64}$

3. $P(\text{both red}), \text{ without replacement}$

ANSWER $\frac{3}{28}$

4. $P(\text{both blue}), \text{ without replacement}$

ANSWER $\frac{5}{14}$

5. $P(\text{one of each}), \text{ without replacement}$

ANSWER $\frac{15}{28}$

6. $P(\text{walks}) \text{ from the table}$

ANSWER $\frac{21}{40}$

7. $P(\text{a boy given the pupil walks})$

ANSWER $\frac{4}{7}$

Hard · stretch and reason

7 PROBLEMS

1. Three tosses: $P(\text{exactly two heads})$

ANSWER $\frac{3}{8}$

2. A bag of 4 white, 6 black; two without replacement: $P(\text{same colour})$

ANSWER $\frac{7}{15}$

3. Same bag: $P(\text{different colours})$

ANSWER $\frac{8}{15}$

4. $P(\text{at least one six in three rolls})$

ANSWER $\frac{91}{216}$

5. A test is 90% accurate; $P(\text{two correct verdicts in two tests})$

ANSWER 0.81

6. Two cards without replacement: $P(\text{both hearts})$

ANSWER $\frac{1}{17}$

7. A machine: $P(\text{fault}) = 0.02$. $P(\text{no fault in 3 items})$

ANSWER ≈ 0.941

07 Homework

Homework — 5 tasks

Draw the tree for every task, and check that the paths total 1.

1. A bag holds 4 red and 6 blue. Two are drawn with replacement. Find $P(\text{both red})$ and $P(\text{one of each})$.
2. The same bag, without replacement. Find $P(\text{both blue})$.

3. Find P (at least one head in three tosses).
4. Draw a two-way table for 30 pupils, 18 of whom are girls and 20 of whom cycle, with 12 girls cycling. Find P (a boy who does not cycle).
5. Two cards are drawn without replacement. Find P (both are kings).

Annual control work

One paper across the whole year: quadratics, systems, trigonometry, progressions and probability.

LESSON 98–99

2 × 40 MIN

ALGEBRA 9, YILLIK NAZORAT ISHI

IGX 1–15 REVIEW

LESSON CLOCK

3

45'

12'

15'

5'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis by chapter	15 min
What to revise	5 min

LEARNING OBJECTIVES

- Work accurately across five chapters under a single time limit.
- Choose the right method quickly when the question does not name it.
- Show the working that carries the method marks.
- Diagnose each lost mark by chapter, not only by question.

TEXTBOOK

National	pp. 1–240
Cambridge	Core & Extended, full course
Workbook	Annual paper A8

01

Explanation

The paper — 50 marks, 45 minutes

Q	TASK	MARKS	CHAPTER
1	Solve $x^2 - 5x + 6 < 0$	5	I
2	Find the domain of $y = \sqrt{x^2 - 9}$	5	I
3	Solve $\{ x + y = 7 ; x^2 + y^2 = 25 \}$	6	II
4	Prove $x^2 + 36 \geq 12x$ and name the equality case	5	II
5	Given $\sin \alpha = \frac{3}{5}$ (II), find $\cos \alpha$ and $\sin 2\alpha$	6	III
6	Evaluate $\cos 210^\circ + \sin 150^\circ$	5	III
7	In an AP, $a_3 = 10$, $a_8 = 30$: find S_{10}	6	IV
8	Sum $16 + 8 + 4 + \dots$ to infinity	5	IV
9	Two dice: find $P(\text{sum } 8)$ and $P(\text{at least one six})$	7	V

METHOD MARKS ARE GIVEN FOR THE METHOD

Q3 carries a mark for choosing substitution; Q5 two for the sign; Q7 one for using the gap of five steps; Q9 two for the complement. Writing the method down, even without finishing, is worth roughly a third of this paper.

Diagnosis by chapter

Fill this in honestly; it is the revision plan for the summer.

CHAPTER	QUESTIONS	MARKS AVAILABLE	IF YOU LOST MARKS, REVISE
I — the quadratic function	1, 2	10	the sign of a trinomial; domains
II — systems and inequalities	3, 4	11	substitution; the square method
III — trigonometry	5, 6	11	the quadrant rule; reduction
IV — progressions	7, 8	11	a_n , S_n , and $ q < 1$
V — probability	9	7	counting, and the complement

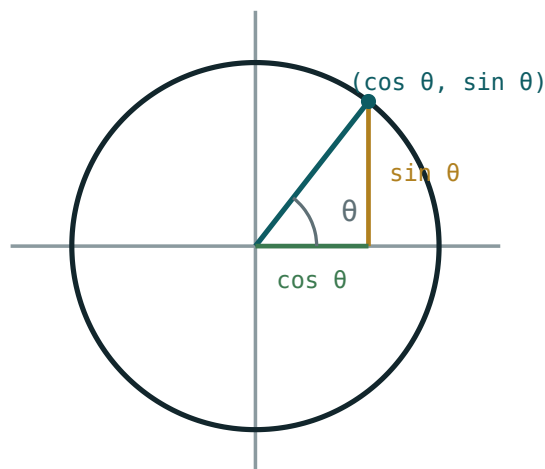
A CHAPTER WITH TWO LOST MARKS IS NOT A WEAK CHAPTER

Look for a whole block that went wrong. Losing four marks spread over five chapters is carelessness under time; losing seven marks in one chapter is a gap in understanding, and only the second needs re-teaching.

What to revise, and in what order

The Grade 10 course opens with powers and roots, then trigonometric equations, then the exponential and logarithmic functions. What matters most from Grade 9 is therefore:

1. **The sign of a quadratic and the method of intervals** — used in every domain question for the next two years.
2. **The quadrant rule and the reduction formulae** — Grade 10 trigonometric equations are unreadable without them.
3. **The addition and double-angle formulae** — they are assumed, not re-taught.
4. **Progressions** — they return in Grade 11 as series and limits.
5. **Counting** — probability in Grade 11 is combinatorics with a division at the end.



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

The single most useful picture of the year, and the one to redraw first in Grade 10.

TWO HOURS A WEEK, NOT TWO DAYS IN AUGUST

Anything practised for twenty minutes a week over the summer survives; anything crammed in the last week before September does not. The five items above, one a week, is enough.

02 Worked examples

EXAMPLE 1 Model answer, Q3: solve $\{x + y = 7; x^2 + y^2 = 25\}$.

- ① One equation is linear — substitute $y = 7 - x$.

The method mark.

② $x^2 + (7 - x)^2 = 25 \Rightarrow 2x^2 - 14x + 24 = 0$

③ $x^2 - 7x + 12 = 0 \Rightarrow x = 3, 4$

- ④ $(3, 4)$ and $(4, 3)$ — both checked.

ANSWER

$(3, 4)$ and $(4, 3)$

EXAMPLE 2 Model answer, Q5: $\sin \alpha = \frac{3}{5}$ in quadrant II.

① $\cos^2 \alpha = 1 - \frac{9}{25} = \frac{16}{25}$

② Quadrant II $\Rightarrow \cos \alpha = -\frac{4}{5}$.

The sign.

③ $\sin 2\alpha = 2 \cdot \frac{3}{5} \cdot \left(-\frac{4}{5}\right)$

④ $= -\frac{24}{25}$

ANSWER

$\cos \alpha = -\frac{4}{5}, \sin 2\alpha = -\frac{24}{25}$

EXAMPLE 3 Model answer, Q9: two dice — $P(\text{sum } 8)$ and $P(\text{at least one six})$.

① $n = 36$.

② Sum 8: $(2,6),(3,5),(4,4),(5,3),(6,2)$ — five pairs.

$$P = \frac{5}{36}.$$

③ At least one six: complement is $5 \times 5 = 25$.

④ $P = 1 - \frac{25}{36} = \frac{11}{36}$

ANSWER

$$\frac{5}{36} \text{ and } \frac{11}{36}$$

03 Interactive model

Return the papers with the chapter column filled in but the marks hidden; pupils diagnose their own weakest chapter before they see their total.

The year in twelve questions

For $a > 0$, $ax^2 + bx + c < 0$ holds:

outside the roots

between them

nowhere

everywhere

The domain of $\sqrt{x - 4}$:

$x > 4$

$x \geq 4$

$x \leq 4$

all x

A system with one linear equation:

substitution

addition

squaring

a graph

To prove $A \geq B$:

test values

show $A - B \geq 0$

draw it

cross-multiply

180° in radians:

$$\frac{\pi}{2}$$

$$\pi$$

$$2\pi$$

$$90$$

$\sin(\alpha + \beta)$ equals:

$$\sin \alpha + \sin \beta$$

$$sc + cs$$

$$cc - ss$$

$$2sc$$

$\cos(\frac{\pi}{2} + \alpha)$ equals:

$$\sin \alpha$$

$$-\sin \alpha$$

$$\cos \alpha$$

$$-\cos \alpha$$

a_n of an AP:

$$a_1 + nd$$

$$a_1 + (n - 1)d$$

$$nd$$

$$a_1 d$$

b_n of a GP:

$$b_1 q^n$$

$$b_1 q^{n-1}$$

$$b_1 + nq$$

$$nq$$

An infinite GP sums when:

$$q > 0$$

$$|q| < 1$$

$$q > 1$$

always

$P(A) + P(A')$ equals:

0

$\frac{1}{2}$

1

2

“At least one” is best done by:

direct counting

the complement

a tree

guessing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Quadratic inequality	Kvadrat tengsizlik	Квадратное неравенство
System	Sistema	Система
Reduction formula	Keltirish formulasi	Формула приведения
Progression	Progressiya	Прогрессия
Probability	Ehtimollik	Вероятность
Method mark	Usul uchun ball	Балл за метод
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Method marks are earned by:

the answer

the working

speed

neatness

Q5 loses marks most often for:

arithmetic

the sign

the formula

the units

Q8 must state:

n

$|q| < 1$

b_1

a decimal

Q9 is quickest by:

listing 36 cases

the complement

a tree

a formula

Seven marks lost in one chapter means:

carelessness

a gap in understanding

bad luck

nothing

The best summer plan is:

two days in August

twenty minutes a week

nothing

the whole textbook

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 - 5x + 6 < 0$

ANSWER $(2, 3)$

2. Domain of $y = \sqrt{x^2 - 9}$

ANSWER $(-\infty, -3] \cup [3, +\infty)$

3. Prove $x^2 + 36 \geq 12x$

ANSWER $(x - 6)^2 \geq 0$

4. $\cos 210^\circ$

ANSWER $-\frac{\sqrt{3}}{2}$

5. $\sin 150^\circ$

ANSWER $\frac{1}{2}$

6. $16 + 8 + 4 + \dots$

ANSWER 32

7. Two dice: $P(\text{sum } 8)$

ANSWER $\frac{5}{36}$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ x + y = 7 ; x^2 + y^2 = 25 \}$

ANSWER $(3, 4), (4, 3)$

2. $\sin \alpha = \frac{3}{5}$ (II): $\cos \alpha$

ANSWER $-\frac{4}{5}$

3. Same: $\sin 2\alpha$

ANSWER $-\frac{24}{25}$

4. $\cos 210^\circ + \sin 150^\circ$

ANSWER $\frac{1 - \sqrt{3}}{2}$

5. $a_3 = 10, a_8 = 30$: d

ANSWER 4

6. Same: S_{10}

ANSWER 200

7. Two dice: $P(\text{at least one six})$

ANSWER $\frac{11}{36}$
Hard · stretch and reason

7 PROBLEMS

1. Solve $\{x^2 - 4 > 0; x^2 - 49 < 0\}$

ANSWER $(-7, -2) \cup (2, 7)$

2. Prove $a + \frac{1}{a} \geq 2$ for $a > 0$

ANSWER $(\sqrt{a} - \frac{1}{\sqrt{a}})^2 \geq 0$

3. Evaluate $\sin 225^\circ + \cos 315^\circ + \tan 225^\circ$

ANSWER 1

4. $S_5 = 55$ and $d = 3$ in an AP: a_1

ANSWER 5

5. A GP with $b_1 = 9$ and $S = \frac{27}{2}$: find q

ANSWER $\frac{1}{3}$

6. Three coins: $P(\text{exactly two heads})$

ANSWER $\frac{3}{8}$

7. A bag of 3 red, 5 blue; two without replacement: $P(\text{one of each})$

ANSWER $\frac{15}{28}$

07 Homework

Homework — 5 tasks

Do the five tasks from the chapter in which you lost most marks, not one from each.

1. Solve $x^2 - 7x + 10 \leq 0$ and give the answer in set notation.
2. Given $\cos \alpha = -\frac{12}{13}$ in quadrant III, find $\sin 2\alpha$.

3. In an AP, $a_4 = 11$ and $a_9 = 26$. Find S_{12} .
4. Sum $24 + 8 + \frac{8}{3} + \dots$ to infinity.
5. Two dice are rolled. Find $P(\text{sum less than } 5)$.

Annual revision — the year on one page, and what Grade 10 expects

Five chapters, five sentences, and an honest look at what has to be secure by September.

LESSON 100–102

3 × 40 MIN

ALGEBRA 9, YILLIK TAKRORLASH

IGX FULL COURSE

LESSON CLOCK

25'

35'

35'

20'

5

Five chapters, five sentences	25 min
Which chapter is this?	35 min
Mixed practice	35 min
What Grade 10 expects	20 min
The summer plan	5 min

LEARNING OBJECTIVES

- State the central idea of each of the five chapters in one sentence.
- Recognise which chapter a question belongs to before choosing a method.
- Identify personally weak areas and plan summer practice around them.
- See how each chapter is used in Grade 10 and beyond.

TEXTBOOK

National	pp. 1–240
Cambridge	Core & Extended, full course
Workbook	Revision exercises

01

Explanation

Five chapters, five sentences

CHAPTER	THE SENTENCE	THE PICTURE
I — the quadratic function	a gives the direction, D the number of roots, and the roots give the sign	a parabola
II — systems and inequalities	solve each line separately, then combine — and to prove, move everything to one side	a number line
III — trigonometry	an angle is a point on the unit circle, and every formula is a statement about that point	the unit circle
IV — progressions	add the same thing, or multiply by the same thing — and both sums have a trick	a straight line, or a curve
V — probability	count what can happen, count what you want, divide	a tree

FIVE PICTURES ARE ENOUGH

A parabola, a number line, the unit circle, a straight line of terms, and a tree. If those five can be drawn from memory with their labels, the year is secure — and anything forgotten can be rebuilt from them.

Which chapter is this?

The hardest part of a mixed paper is not the mathematics but the recognition.

Practise it directly.

QUESTION	CHAPTER	BECAUSE
“Find the domain of $\sqrt{x^2 - 4}$ ”	I	a quadratic must be ≥ 0
“Prove $a^2 + b^2 \geq 2ab$ ”	II	one side, then a square
“Evaluate $\cos 750^\circ$ ”	III	strip turns, then reduce
“How many multiples of 7 below 500?”	IV	an AP with $d = 7$
“ $P(\text{at least one six})$ ”	V	the complement
“Two numbers with sum 9 and product 20”	I or II	Vieta, or a symmetric system

READ THE WHOLE QUESTION BEFORE CHOOSING A METHOD

The last row has two correct routes and one wrong reflex — trying to solve it as a pair of simultaneous linear equations. Thirty seconds of reading saves ten minutes of the wrong algebra.

Mixed practice

Work these without knowing which chapter each belongs to; that is the point.

TASK	ANSWER
Solve $x^2 - x - 12 \geq 0$	$(-\infty, -3] \cup [4, +\infty)$
Solve $\{ x - y = 1 ; xy = 6 \}$	$(3, 2), (-2, -3)$
Evaluate $\sin 120^\circ \cos 30^\circ$	$\frac{3}{4}$
S_{20} of 1, 5, 9, ...	780
Sum $27 + 9 + 3 + \dots$	$\frac{81}{2}$
$P(\text{a card is a red king})$	$\frac{1}{26}$

MIXED PRACTICE IS THE ONLY PRACTICE THAT TESTS RECOGNITION

Twenty questions from one chapter train the method; twenty from five chapters train the choice — and it is the choice that a mixed examination actually measures.

What Grade 10 expects

GRADE 10 TOPIC	NEEDS, FROM GRADE 9
powers with rational exponents	surds and index laws
irrational equations	domains, and checking for false roots
trigonometric equations	the unit circle, quadrants, reduction formulae
the exponential function	geometric progressions
logarithms	index laws, and the idea of an inverse
probability (Grade 11)	counting, and the complement

Cambridge continues in parallel: Grade 9 covered IGCSE chapters up to trigonometry and sequences; Grade 10 takes the remaining Extended chapters and the first Pure Mathematics topics. The two courses have run together all year, and they continue to.

THE SUMMER PLAN, IN ONE LINE

One page a week: the parabola and the sign rule; the unit circle with the exact values; the four progression formulae; ten mixed problems. Four short sessions across the summer, and September starts from a full page rather than a blank one.

02 Worked examples

EXAMPLE 1 Solve $\{x - y = 1; xy = 6\}$.

- ① One equation is linear: $x = y + 1$.
Chapter II.
- ② $(y + 1)y = 6 \Rightarrow y^2 + y - 6 = 0$
- ③ $y = 2$ or $y = -3$.
- ④ $(3, 2)$ and $(-2, -3)$ — both checked.

ANSWER

$(3, 2)$ and $(-2, -3)$

EXAMPLE 2 Evaluate $\sin 120^\circ \cos 30^\circ$.

- ① $\sin 120^\circ = \sin(180^\circ - 60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$
Chapter III.
- ② $\cos 30^\circ = \frac{\sqrt{3}}{2}$
- ③ $\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}$
- ④ $= \frac{3}{4}$

ANSWER

$\frac{3}{4}$

EXAMPLE 3 Find S_{20} for 1, 5, 9, ..., and sum $27 + 9 + 3 + \dots$ to infinity.

① $a_1 = 1, d = 4: a_{20} = 1 + 19 \cdot 4 = 77.$

Chapter IV.

② $S_{20} = \frac{20(1 + 77)}{2} = 780$

③ $b_1 = 27, q = \frac{1}{3}, \text{ and } |q| < 1.$

④ $S = \frac{27}{1 - \frac{1}{3}} = \frac{81}{2}$

ANSWER

780 and $\frac{81}{2}$

03 Interactive model

Give the class twelve questions in random order and ask only for the chapter of each — no solving. Recognition, practised alone, is what a mixed paper rewards.

Which chapter, and which method?

$x^2 - 5x + 6 = 0$ belongs to:

quadratics

sequences

probability

trigonometry

Its roots are:

2 and 3

-2 and -3

1 and 6

-1 and -6

The discriminant of $x^2 + 2x + 5$ is:

-16

16

24

4

3, 7, 11, ... has a_{10} equal to:

39

40

43

37

A GP with $a_1 = 2$ and $q = \frac{1}{2}$ has infinite sum:

2

4

1

no sum

$\sin^2\alpha + \cos^2\alpha$ equals:

0

1

2

α

$\sin 150^\circ$ equals:

-0.5

0.5

$\frac{\sqrt{3}}{2}$

1

One die: $P(\text{a prime})$ is:

$\frac{1}{6}$

$\frac{1}{3}$

$\frac{1}{2}$

$\frac{2}{3}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Summary	Xulosa	Итог
Concept map	Tushunchalar xaritasi	Карта понятий
Prerequisite	Zaruriy bilim	Необходимое условие
Consolidation	Mustahkamlash	Закрепление
Weak area	Zaif tomon	Слабое место
Practice	Mashq	Практика
Preparation	Tayyorgarlik	Подготовка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Find the domain of $\sqrt{x^2 - 4}$ ” belongs to:

Chapter I

Chapter II

Chapter III

Chapter V

“Evaluate $\cos 750^\circ$ ” belongs to:

Chapter I

Chapter II

Chapter III

Chapter IV

“How many multiples of 7 below 500” belongs to:

Chapter II

Chapter III

Chapter IV

Chapter V

“ P (at least one six)” is done by:

listing

the complement

a formula

a graph

Grade 10 trigonometric equations most need:

progressions

the unit circle and reduction

probability

Vieta

The best revision is:

one chapter at a time

mixed questions

reading the book

copying notes

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 - x - 12 \geq 0$

ANSWER $(-\infty, -3] \cup [4, +\infty)$

2. Domain of $\sqrt{x^2 - 4}$

ANSWER $(-\infty, -2] \cup [2, +\infty)$

3. $\sin 120^\circ$

ANSWER $\frac{\sqrt{3}}{2}$

4. $\cos 30^\circ$

ANSWER $\frac{\sqrt{3}}{2}$

5. S_{20} of 1, 5, 9, ...

ANSWER 780

6. $27 + 9 + 3 + \dots$

ANSWER $\frac{81}{2}$

7. $P(\text{a card is a red king})$

ANSWER $\frac{1}{26}$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ x - y = 1 ; xy = 6 \}$

ANSWER $(3, 2), (-2, -3)$

2. $\sin 120^\circ \cos 30^\circ$

ANSWER $\frac{3}{4}$

3. Prove $a^2 + b^2 \geq 2ab$

ANSWER $(a - b)^2 \geq 0$

4. Multiples of 7 below 500

ANSWER 71

5. $\cos 750^\circ$

ANSWER $\frac{\sqrt{3}}{2}$

6. Two numbers: sum 9, product 20

ANSWER 4 and 5

7. Two dice: $P(\text{at least one six})$

ANSWER $\frac{11}{36}$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\{ x^2 + y^2 = 20 ; xy = 8 \}$

ANSWER $(4, 2), (2, 4), (-4, -2), (-2, -4)$

2. Prove $\frac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$

ANSWER Use $2\cos^2\alpha$

3. An AP has $S_{10} = 155$ and $a_1 = 2$: find d

ANSWER 3

4. A GP has $S = 12$ and $b_1 = 8$: find q

ANSWER $\frac{1}{3}$

5. Three coins: $P(\text{at least two heads})$

ANSWER $\frac{1}{2}$

6. Solve $\sin 3x + \sin x = 0$

ANSWER $x = \frac{\pi k}{2}$

7. Domain of $y = \frac{\sqrt{x+1}}{x-2}$

ANSWER $[-1, 2) \cup (2, +\infty)$

07 Homework

Summer work — 5 tasks, one a week

Not five tasks for one evening: one page a week, spread across the summer.

1. Week 1: draw the parabola with its sign rule and solve ten quadratic inequalities.
2. Week 2: draw the unit circle with all eight exact values, from memory.
3. Week 3: write the four progression formulae and do ten mixed progression problems.

4. Week 4: do ten probability problems, using the complement wherever “at least” appears.
5. Week 5: do the annual control work again, from a blank page, and compare.